

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary and Renal

Lecture: CPR 04 Cardiac Cycle

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BREATHE WITH THE HEART



DE STRESS
MONDAY

#DESTRESSMONDAY

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Recommended readings

Chapter 13: Mechanics of the Cardiac Pump. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 4: Cardiovascular Physiology: Costanzo Physiology, Seventh Edition.
Linda S. Costanzo PhD.

Slides with blue background are the supplemental information associated with the previous slides and thus, might not be explained during the lecture time. However, please read the slides for clarification as they are equally important to know.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0090 Illustrate and explain, in a correct temporal relationship, the phases of the cardiac cycle (atrial systole, isovolumetric contraction, rapid ventricular ejection, reduced ventricular ejection, isovolumetric relaxation, rapid ventricular filling, reduced ventricular filling/diastasis) and the relationship of each phase with the left ventricular pressure curve, the left ventricular volume curve, the left atrial pressure curve, the aortic pressure curve, and the heart sounds (S1, S2, S3, S4)

SOM.MK.II.BPM1.3.CPR.2.PHYS.0091 Illustrate and explain the ventricular volume-pressure relationship during the different phases of the cardiac cycle in the pressure-volume loop, and the points in the graph that correspond to the mitral valve closure (S1), the mitral valve opening, aortic valve closure (S2), the aortic valve opening, S3 sound, S4 sound, end diastolic volume, and end systolic volume

SOM.MK.II.BPM1.3.CPR.2.PHYS.0092 Compare the timing of the different phases of the cardiac cycle in the left side versus the right side the heart

SOM.MK.II.BPM1.3.CPR.2.PHYS.0093 List the role of factors that contribute to the formation of the turbulent flow

SOM.MK.II.BPM1.3.CPR.2.PHYS.0094 Describe the timing and causes of the four heart sounds

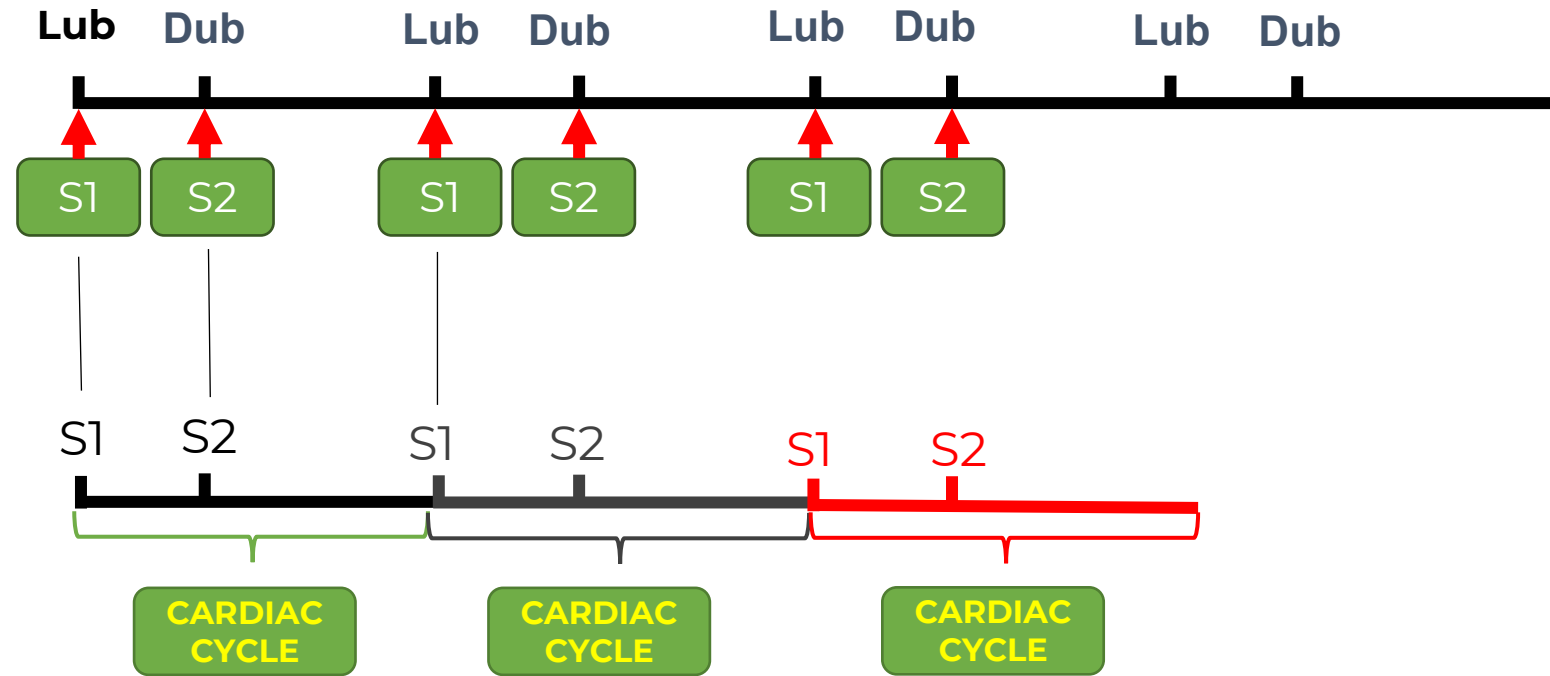
SOM.MK.II.BPM1.3.CPR.2.PHYS.0095 Identify the characteristics of the heart murmurs associated with mitral stenosis, mitral regurgitation, mitral valve prolapse, aortic stenosis, aortic regurgitation, persistent ductus arteriosus, and ventricular septal defect

Lecture Outline

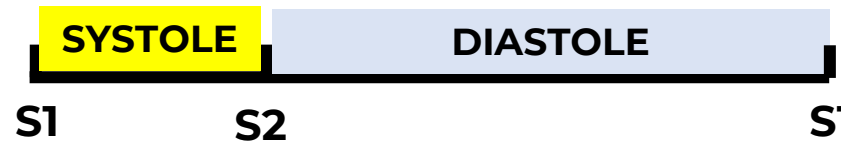
- **Cardiac Cycle Phases**
- **Graphic representation of changes in pressure, volume & sounds**
- **Left Ventricular Pressure-Volume Loop**
- **Left side versus Right side of the heart**

- **Turbulent Flow**
- **Heart Sounds**
- **Heart Murmurs**

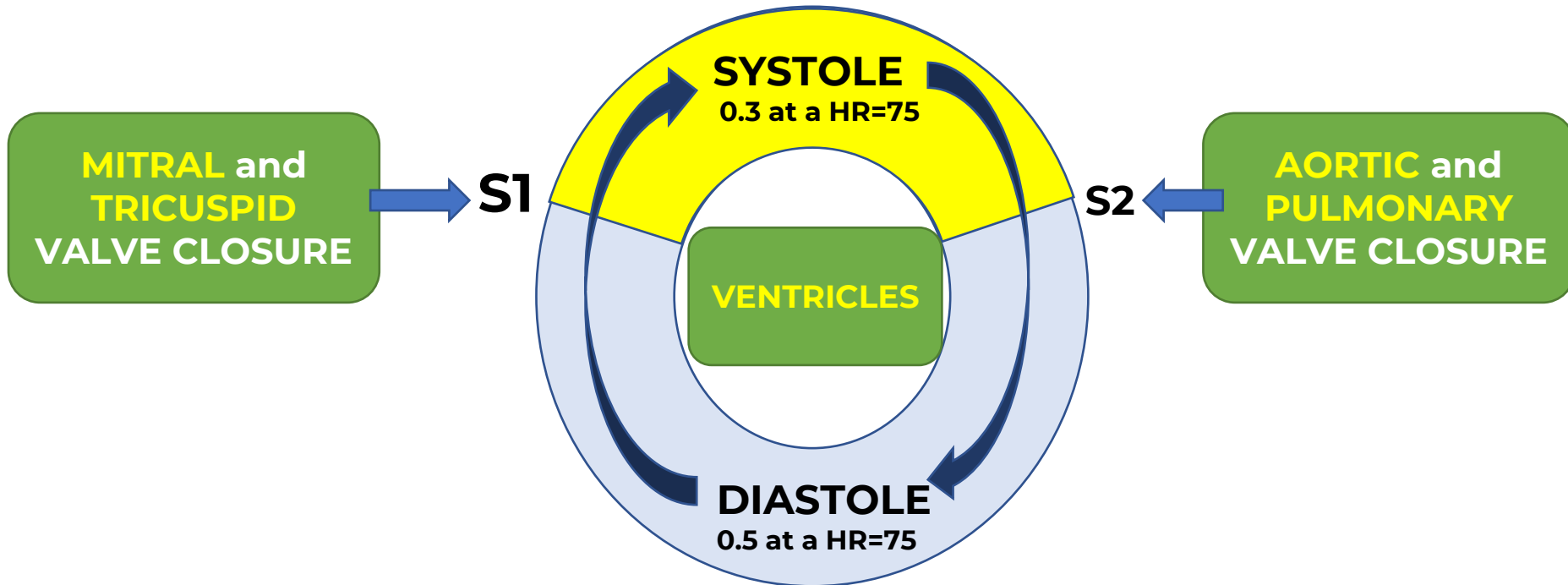
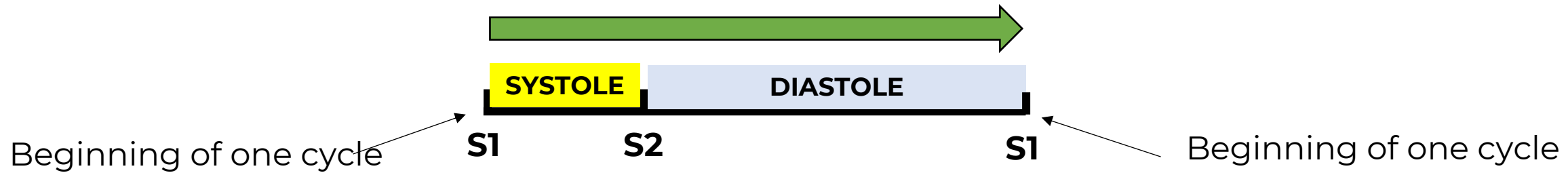
CARDIAC CYCLE: Definition and duration



S1=First heart sound
S2=Second heart sound



CARDIAC CYCLE: Definition and duration



Ventricles spend more time in relaxation than contraction

CARDIAC CYCLE: Definition and duration

*The events that occur in the heart from the beginning of one heartbeat to the beginning of the next are called the **CARDIAC CYCLE**.*

Each cycle is initiated by spontaneous generation of an **Action potential** in the **Sinoatrial node** (pacemaker of the heart)

The cardiac cycle consists of a period of relaxation called **DIASTOLE**, during which the heart fills with blood, followed by a period of contraction called **SYSTOLE**

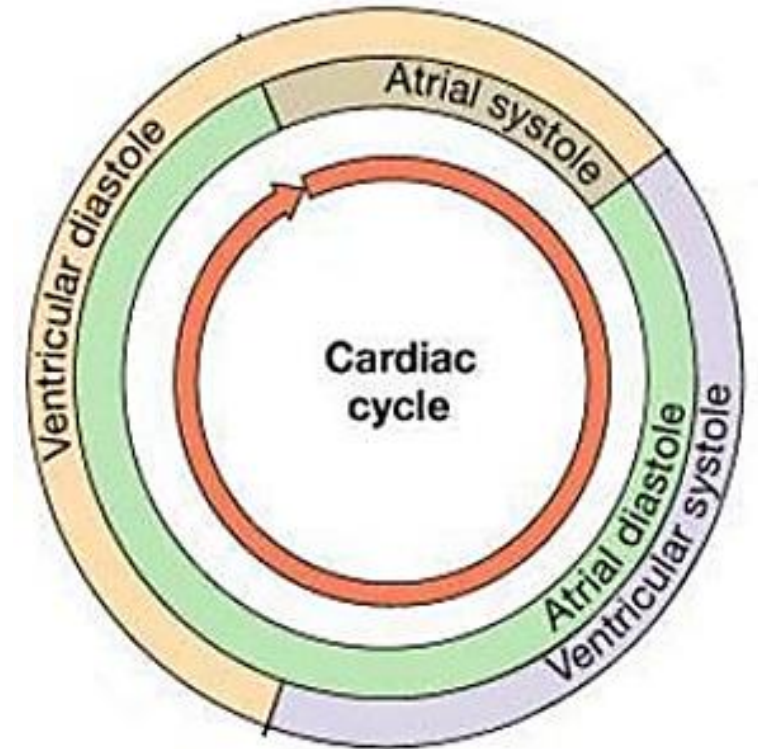
Duration of the Cardiac Cycle: Varies **inversely** with the heart rate

Normal heart rate ranges: 60-90 beats/min

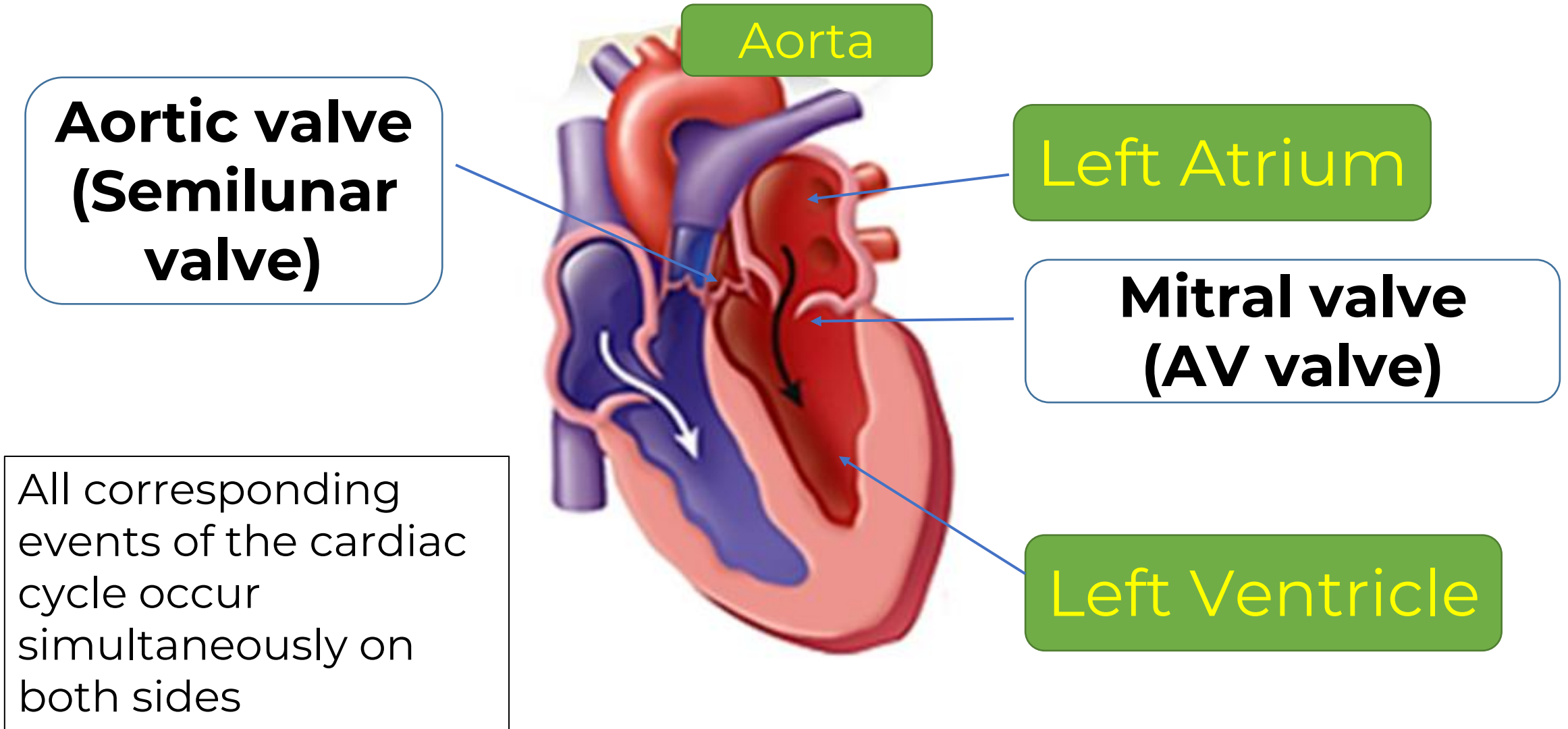
Cardiac cycle ranges: 0.67sec to 1.0 sec.

Phases of the Cardiac Cycle

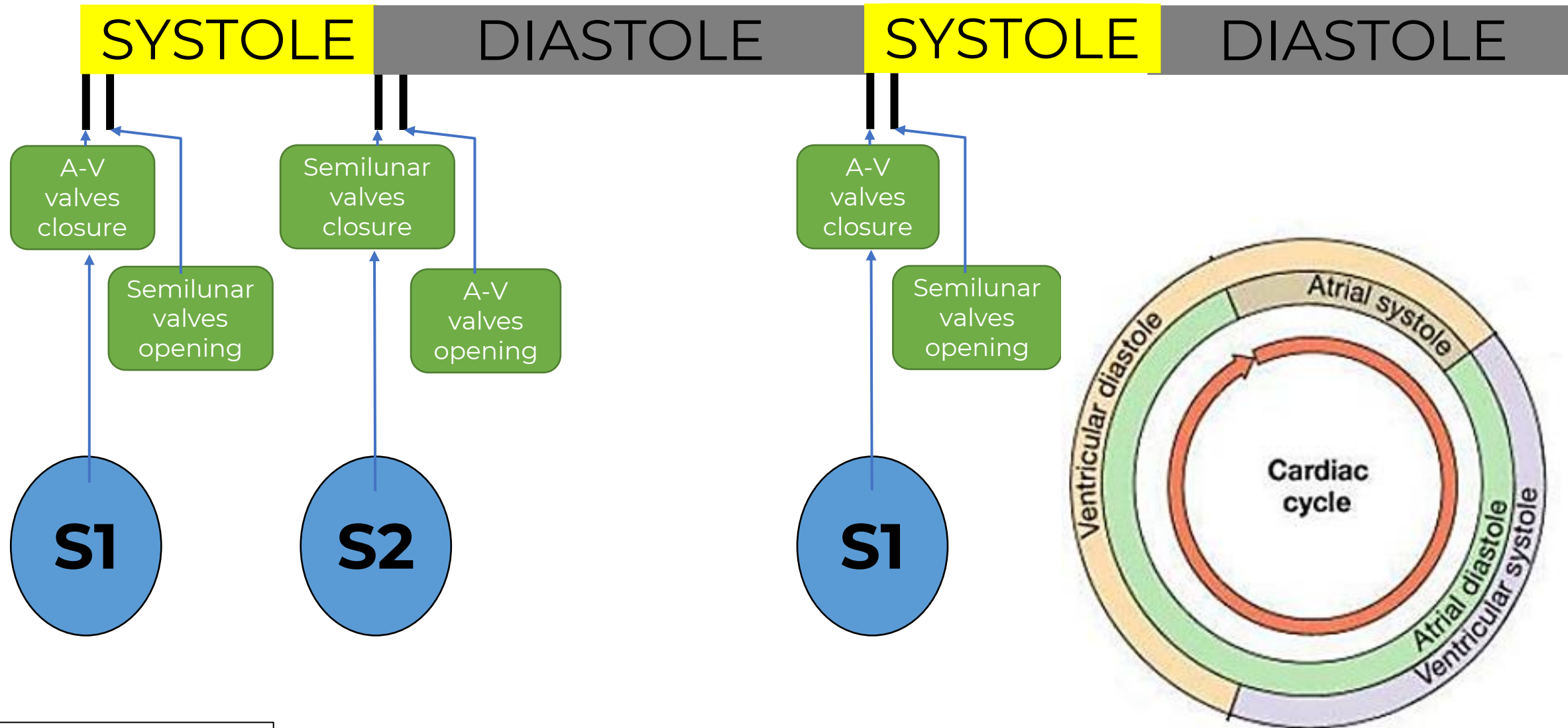
- Sequence of events that occur during a cardiac cycle are called phases of the cardiac cycle
- Ventricular events: Ventricular systole & Ventricular diastole
- Atrial events: Atrial systole & Atrial diastole
- Atrial events *merge into ventricular events* and ventricles being the primary pumps, phases are explained in terms of the **ventricular events**



Phases of the Cardiac Cycle: Left side of the heart



Phases of the Cardiac Cycle



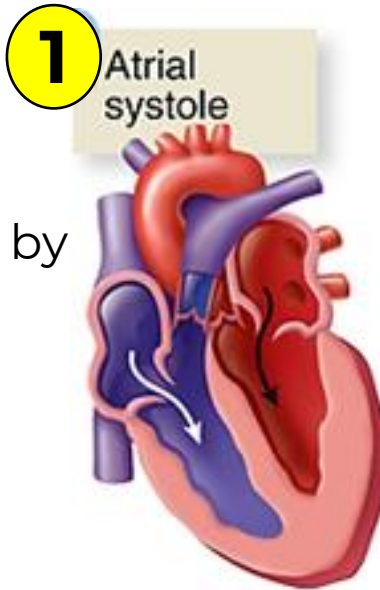
SEVEN PHASES OF THE CARDIAC CYCLE

- 1. Atrial systole**
 - 2. Isovolumetric Ventricular Contraction**
 - 3. Rapid Ventricular Ejection**
 - 4. Reduced Ventricular Ejection**
 - 5. Isovolumetric Ventricular Relaxation**
 - 6. Rapid Ventricular Filling**
 - 7. Reduced Ventricular Filling**
- } Ventricular Diastole
- } Ventricular Systole
- } Ventricular Diastole

Begin here →

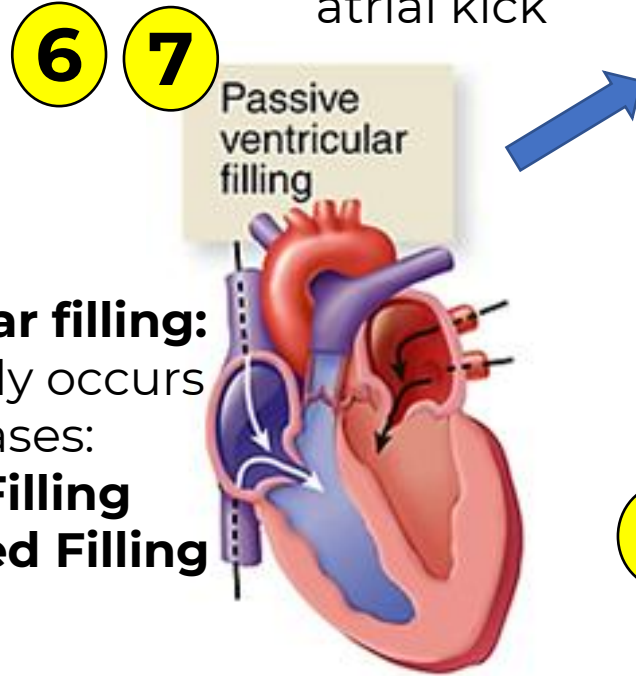
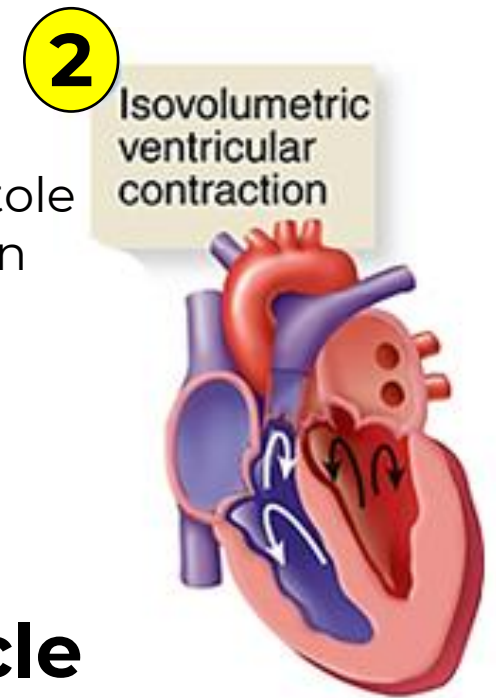
Atrial Systole

- Last part of ventricular filling by atrial kick



Isovolumetric contraction

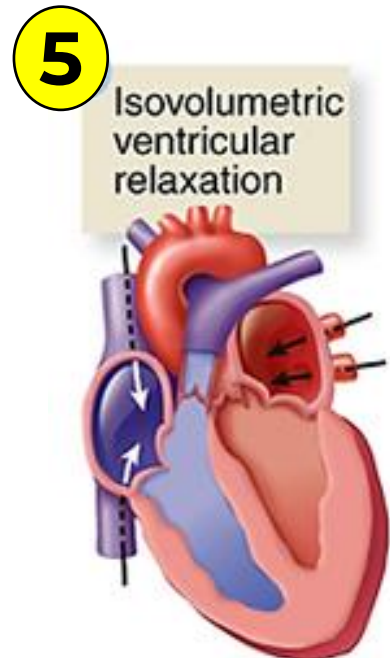
- First part of systole with NO ejection



Ventricular filling:

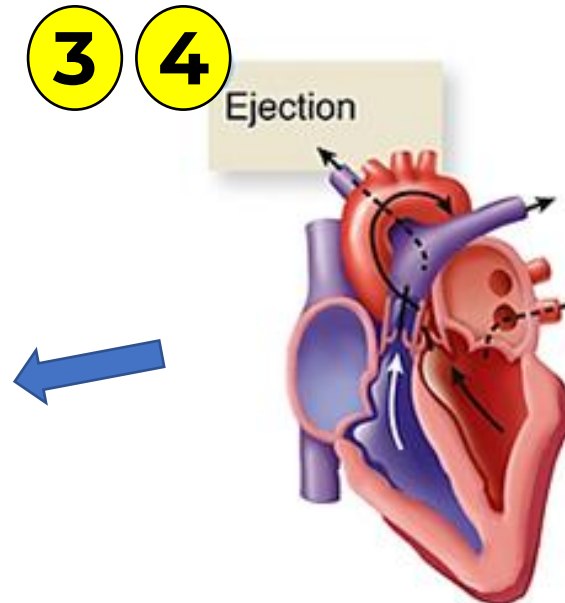
- Passively occurs in 2 phases:
- **Rapid Filling**
- **Reduced Filling**

Phases of cardiac cycle



Isovolumetric relaxation

- First part of diastole with NO filling because both valves are closed



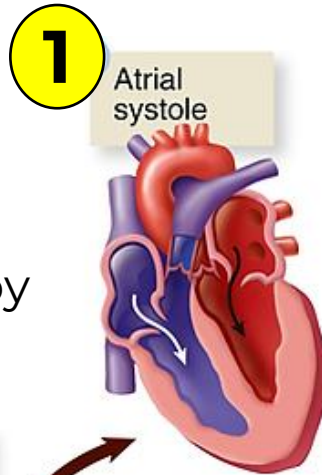
Ventricular ejection:

- Occurs in 2 phases
-
- **Rapid Ejection**
- **Reduced Ejection**

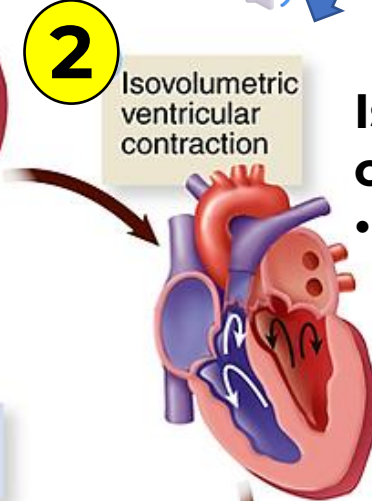
VENTRICULAR DIASTOLE – 4 PHASES

Atrial Systole

- Last part of ventricular filling by atrial kick

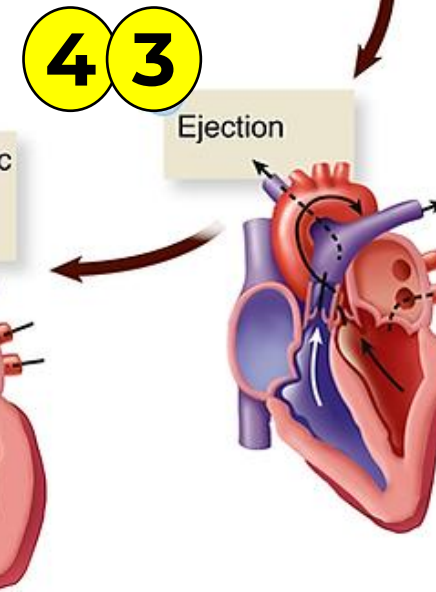


S1
MITRAL and TRICUSPID VALVE CLOSURE



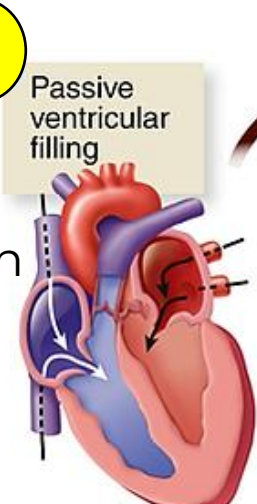
Isovolumetric contraction

- First part of systole with NO ejection



Ventricular ejection:

- Occurs in 2 phases –
- **Rapid Ejection**
- **Reduced Ejection**



Isovolumetric relaxation

- First part of diastole with NO filling because both valves are closed



Ventricular filling:

- Passively occurs in 2 phases:
- **Rapid Filling**
- **Reduced Filling**

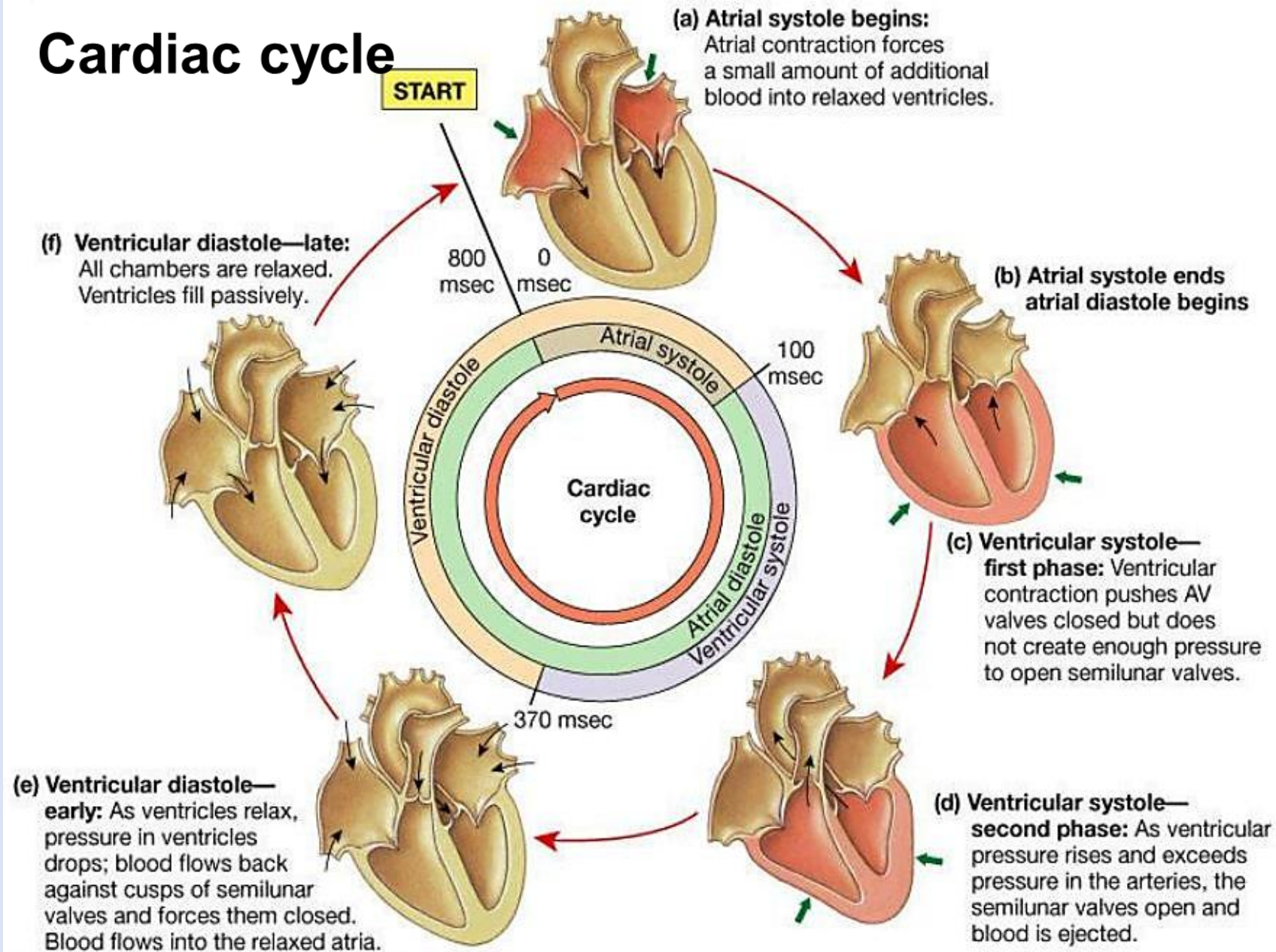


Cardiac cycle

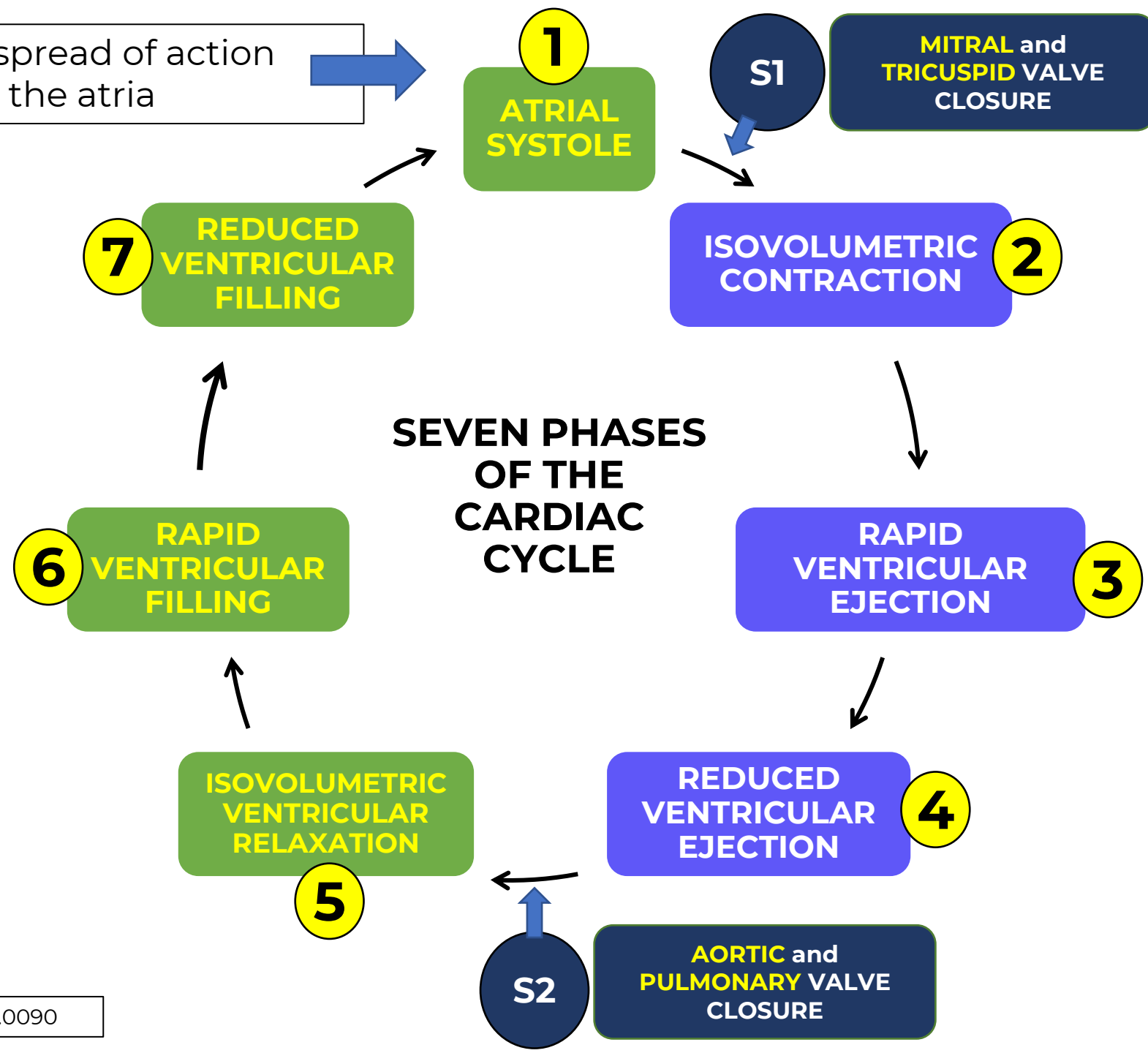
VENTRICULAR SYSTOLE – 3 PHASES

S2
AORTIC and PULMONARY VALVE CLOSURE

Cardiac cycle



Origin and spread of action potential to the atria

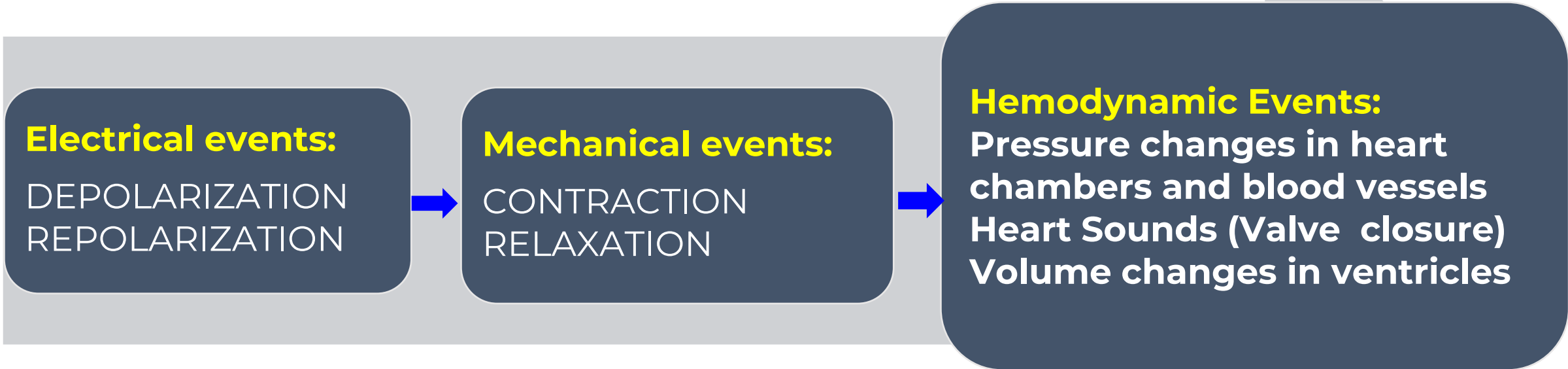


Lecture Outline

- **Cardiac Cycle Phases**
- **Graphic representation of changes in pressure, volume & sounds**
- **Left Ventricular Pressure-Volume Loop**
- **Left side versus Right side of the heart**

- **Turbulent Flow**
- **Heart Sounds**
- **Heart Murmurs**

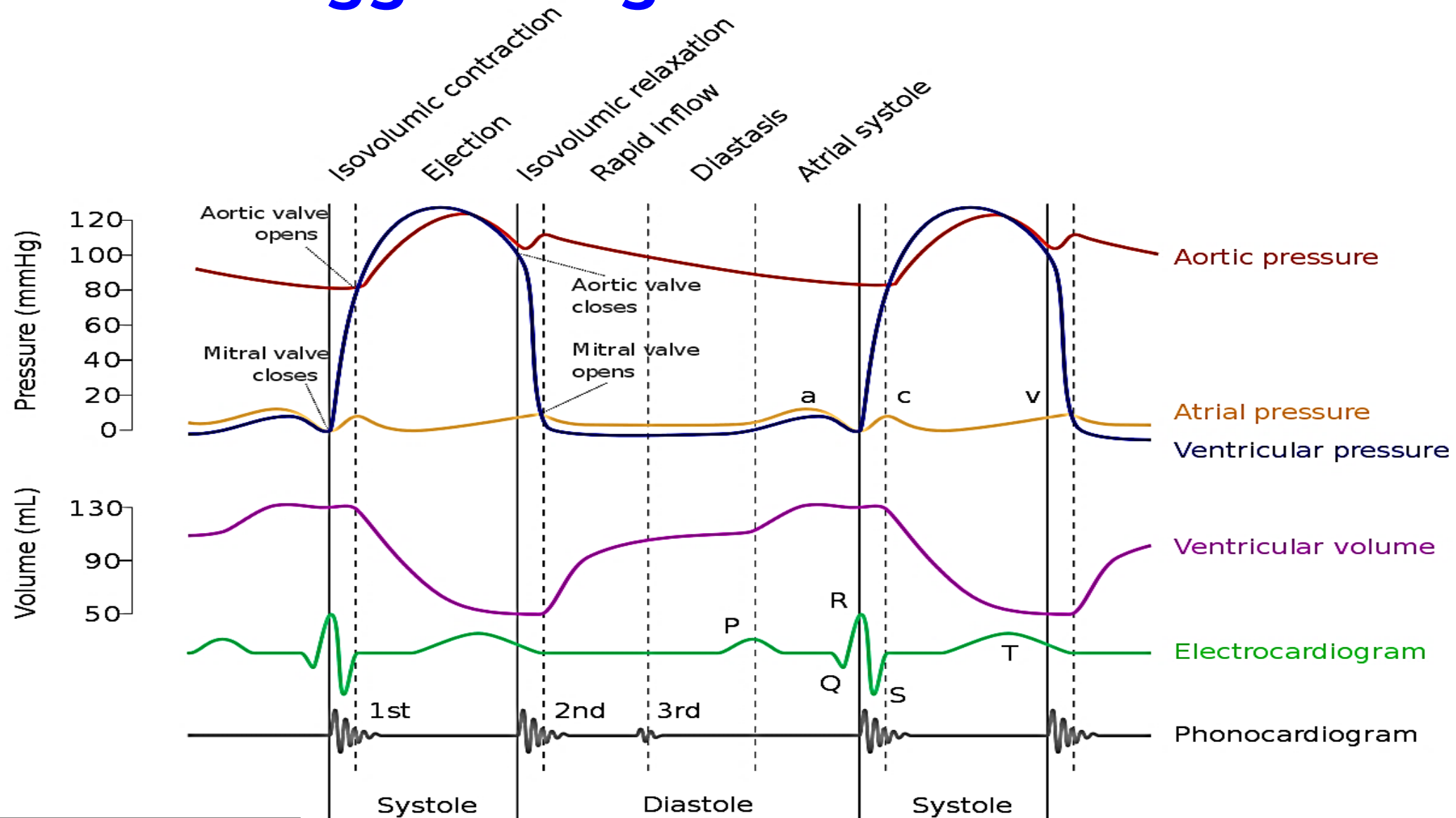
Changes in Pressure, Volume & Sounds during Cardiac Cycle



The changes are recorded using instruments:

- **Electrical activity** **ECG machine**
 - **Pressure changes (atria, ventricles & major arteries)**
 - **Blood Volume changes (mainly in ventricles,** **Echocardiograph**
 - **Sounds** **Phonocardiograph**
- Cardiac Catheterization**

Graphic representation of the changes is also known as **Wiggers diagram**

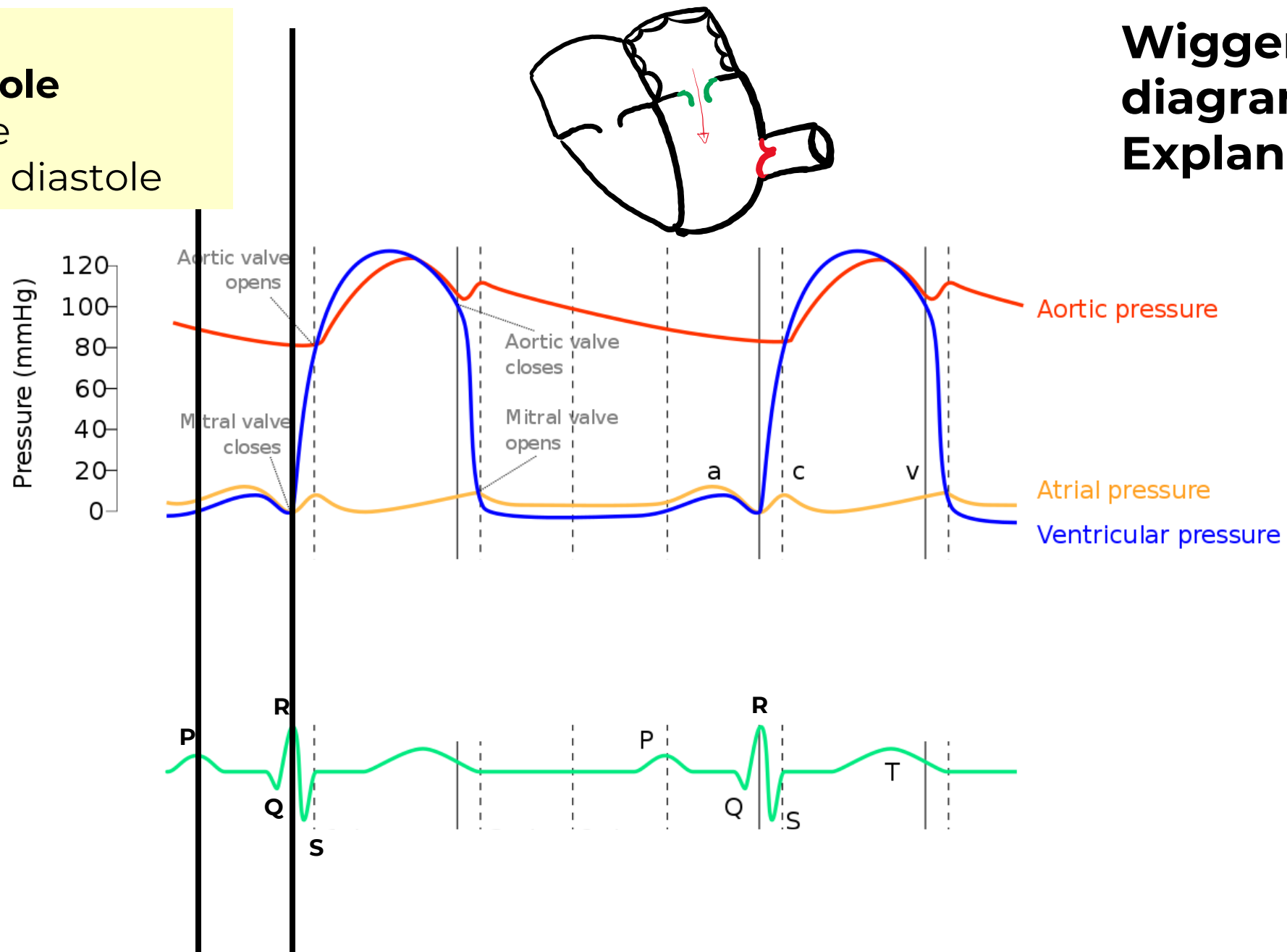


DIASTOLE

Atrial Systole

During late ventricular diastole

Wiggers diagram: Explanation

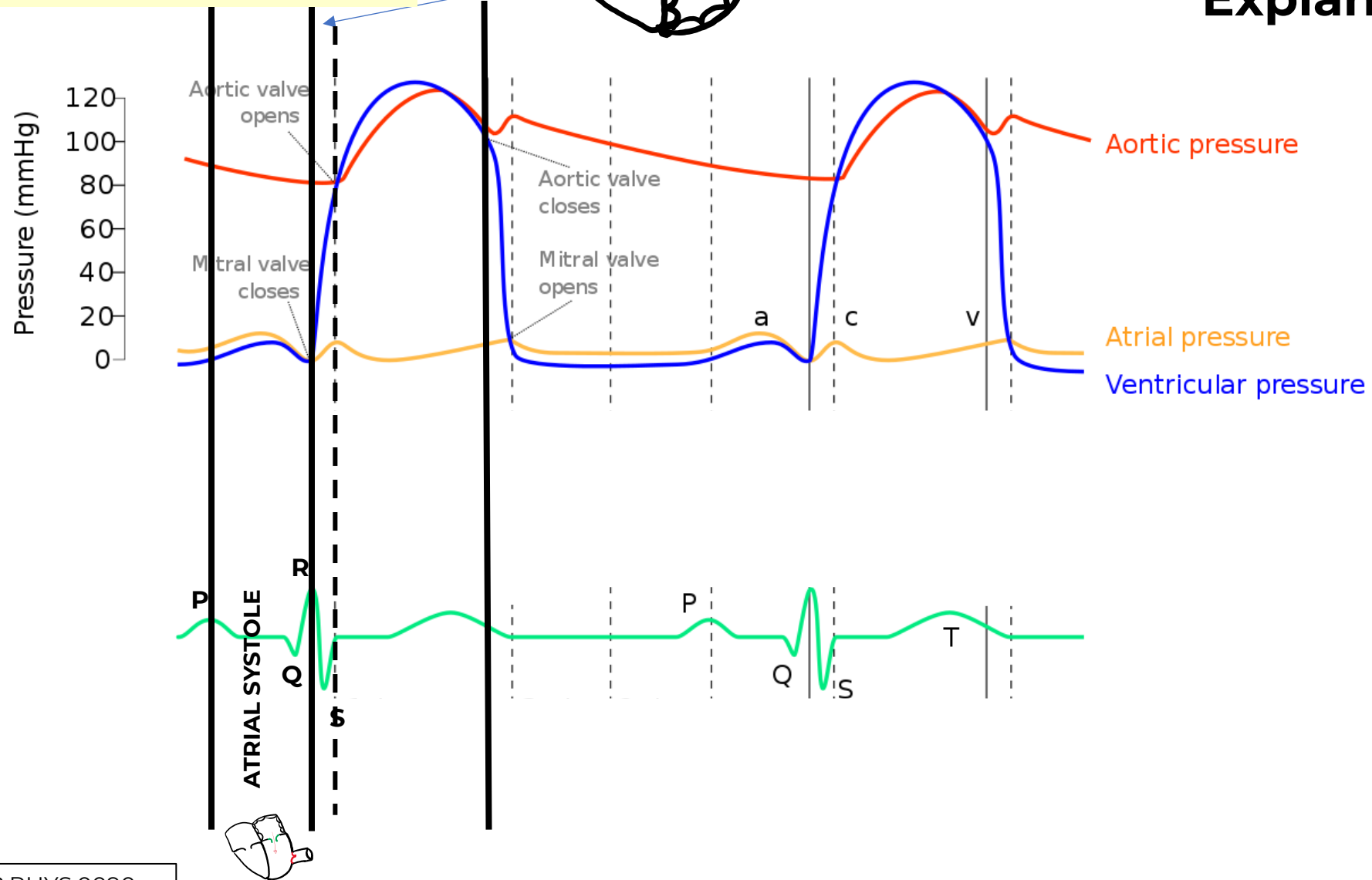


SYSTOLE

Isovolumetric contraction

Beginning: mitral valve closure

Wiggers diagram: Explanation

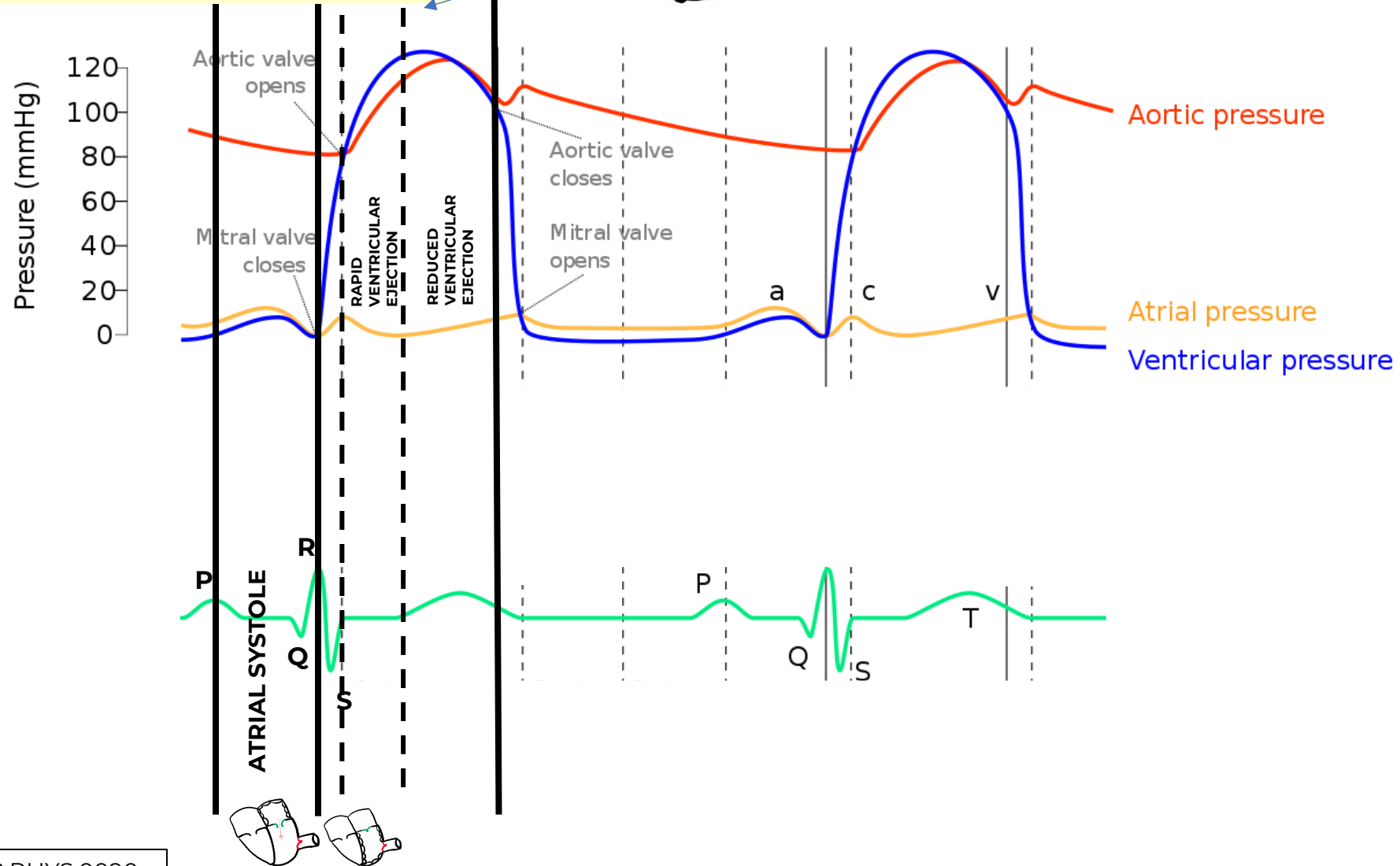


SYSTOLE

Ventricular ejection:

- Beginning: aortic valve opening

Wiggers diagram: Explanation

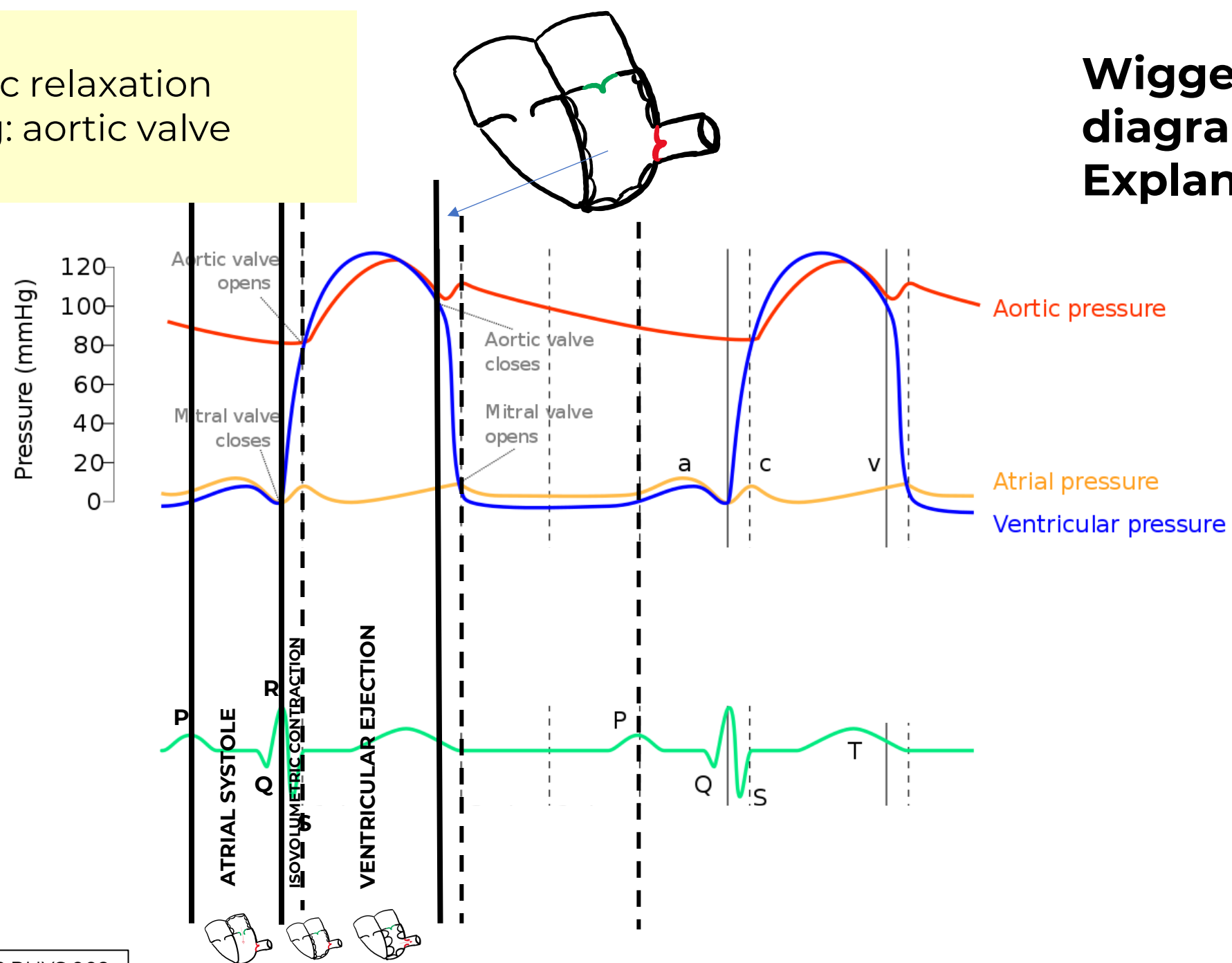


DIASTOLE

Isovolumetric relaxation

- Beginning: aortic valve closure

Wiggers diagram: Explanation

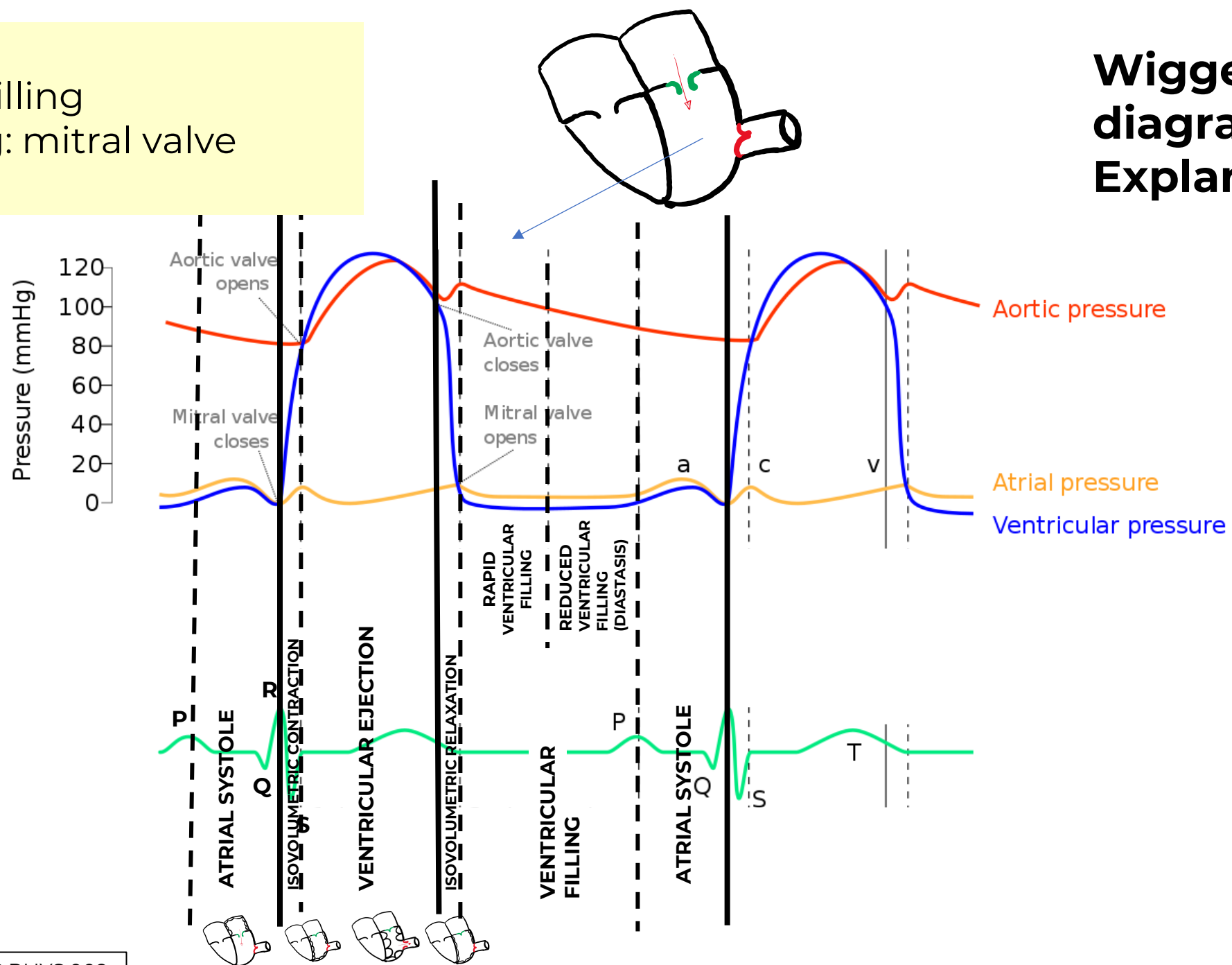


DIASTOLE

Ventricular filling

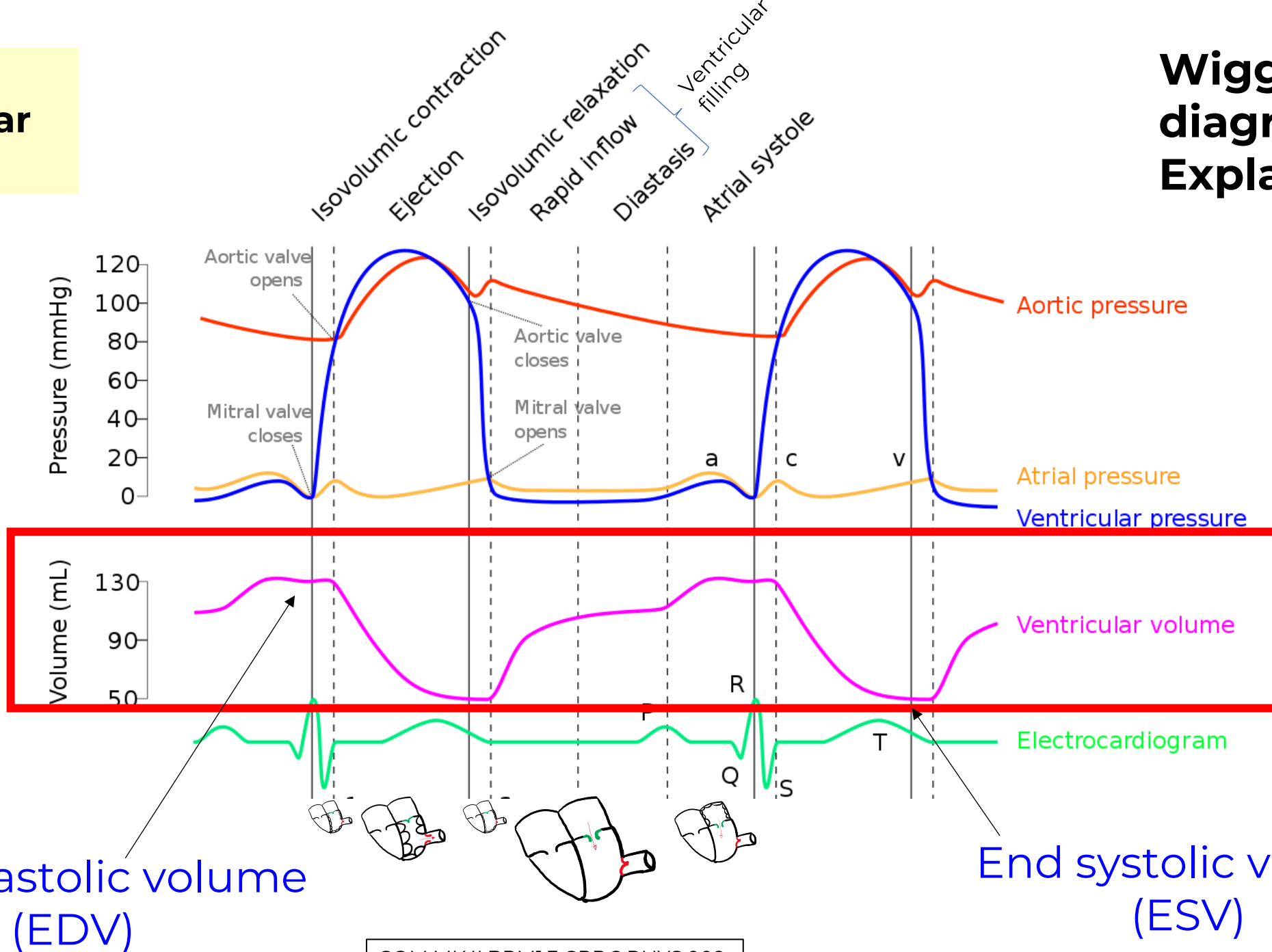
- Beginning: mitral valve opening

Wiggers diagram: Explanation

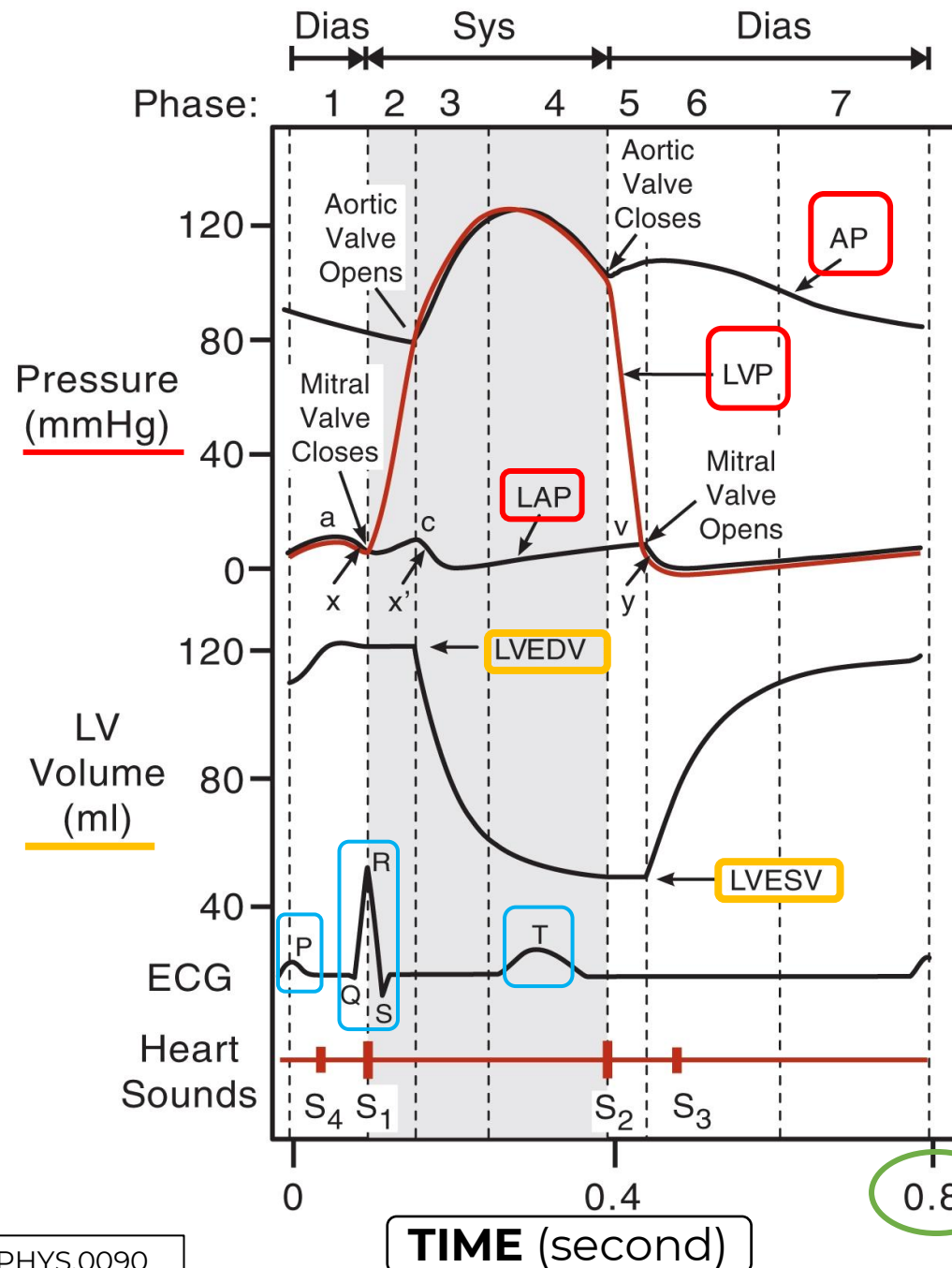


Left Ventricular Volume

Wiggers diagram: Explanation



Wiggers diagram



Pressure (mmHg):

AP: aortic pressure

LVP: left ventricle pressure

LAP: left atrial pressure

Left Ventricle Volume (mL):

LVEDV: left ventricle end diastolic volume

LVESV: LV end systolic volume

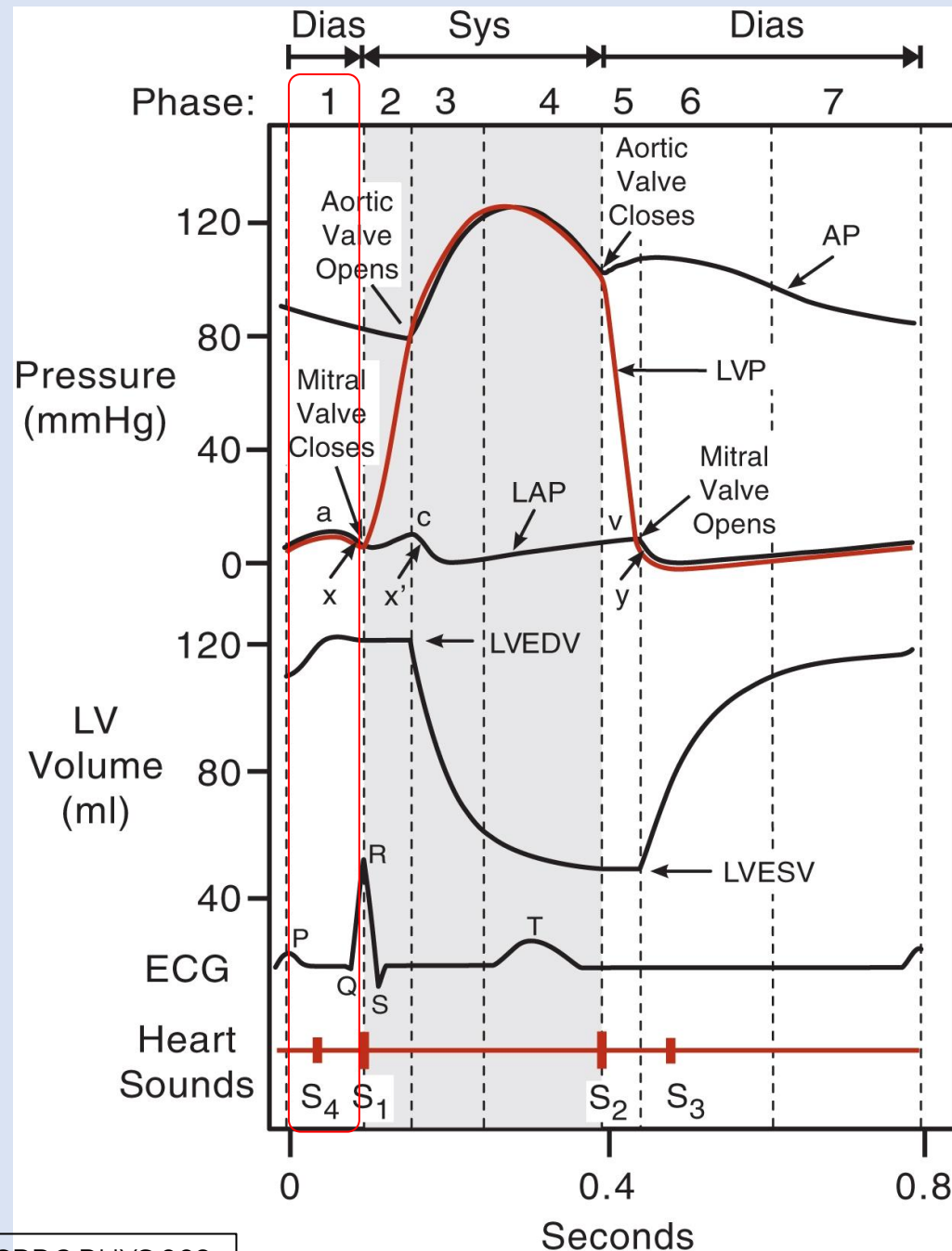
Electrocardiogram:

P wave: atrial depolarization

QRS complex: ventricular depolarization

T wave: ventricular repolarization

Duration of CARDIAC CYCLE with HR=75/min



1. ATRIAL SYSTOLE:

After SA Node fires an Action Potential the wave of depolarization spread through the Atria (Atrial Depolarization): **P wave (ECG)**, followed by **Atrial Contraction**.

LEFT ATRIAL PRESSURE

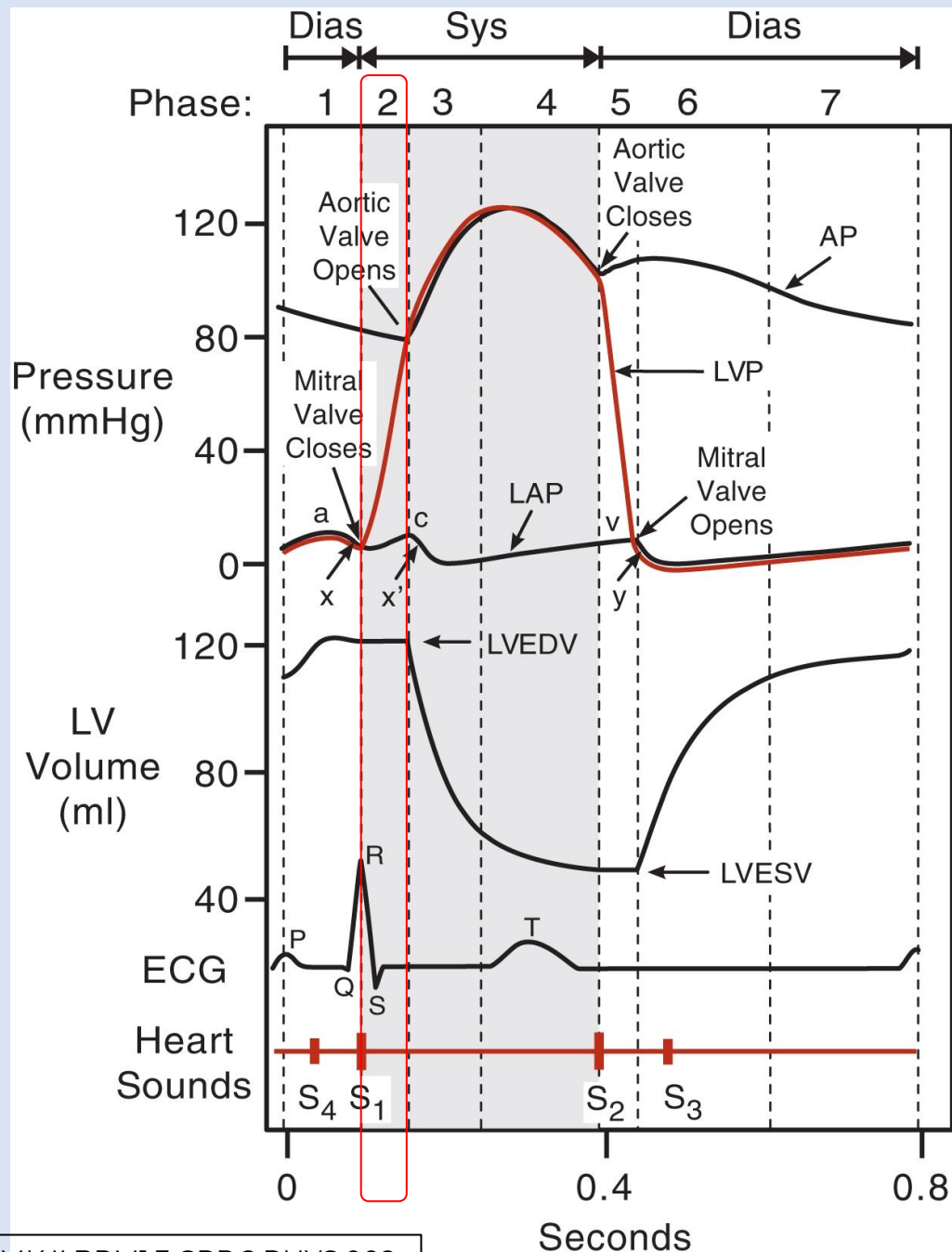
increases and "a" wave is produced in LAP recording.

LEFT VENTRICLE VOLUME

increases by 20% and reaches EDV.

Completion of the Ventricular Filling and ready to contract.

S4 occurs.



2. ISOVOLUMETRIC VENTRICULAR CONTRACTION:

Isovolumetric contraction

Period between mitral valve closing and aortic valve opening

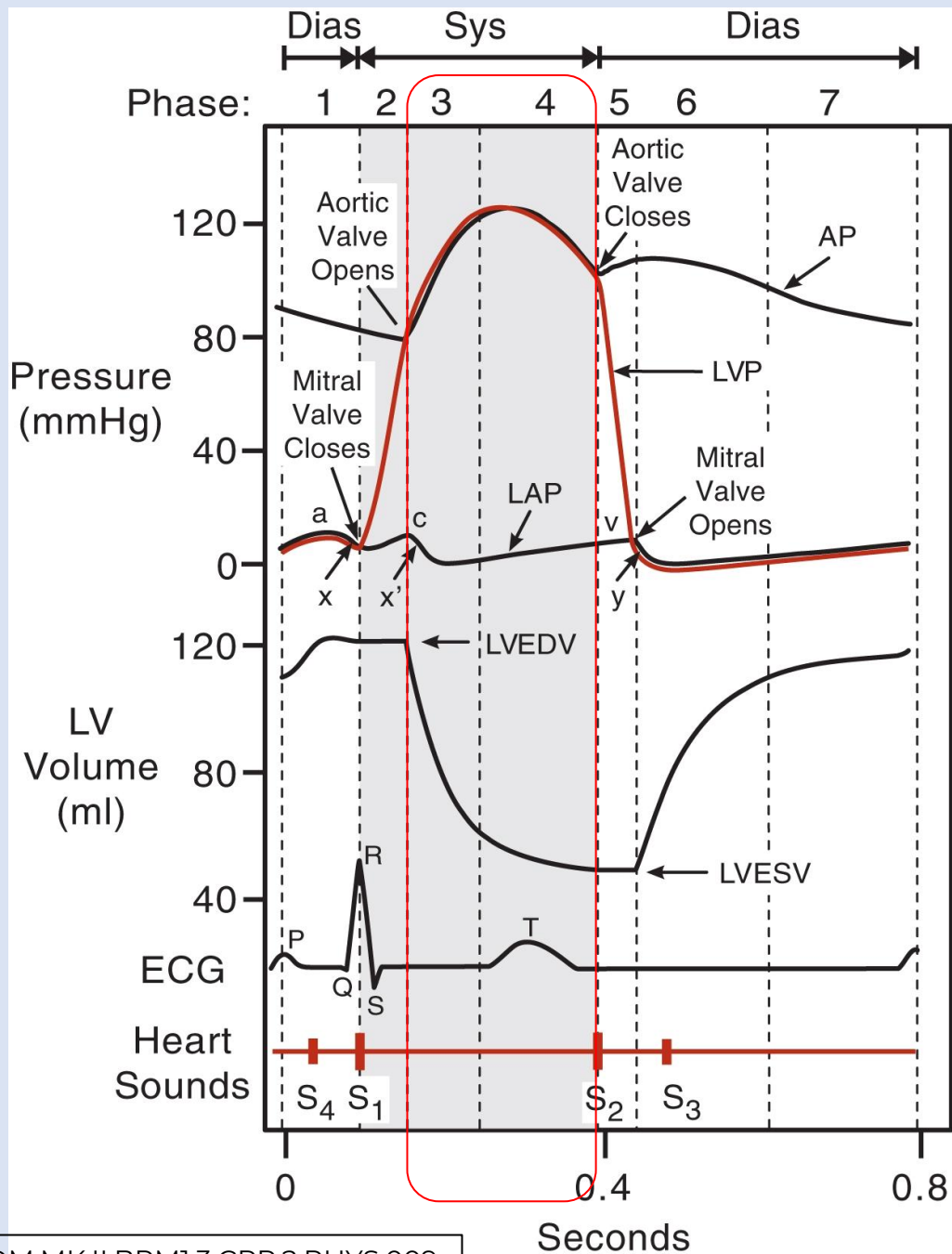
ALL VALVES CLOSED

Depolarization spread through the ventricular myocardium (**QRS complex on ECG**).

Ventricular contraction begins and the pressure starts rising. As soon as the pressure in the ventricles rises above the atrial pressure, the **AV valves close (S1 occurs)**

Contraction against the closed valves makes the LV pressure to rise sharply but the volume remains the same.

During the isovolumetric contraction, the closed mitral valve bulges into the LA, raising its pressure a little. This is seen as the "c" wave in the LA pressure tracing.



3 & 4. RAPID & REDUCED VENTRICULAR EJECTIONS:

Ventricular ejection is the Period between aortic valve opening and closing.

Aortic valve opens at the beginning of this phase when Left Ventricular Pressure rises slightly above the aortic pressure and the ejection starts.

It occurs at 2 phases:

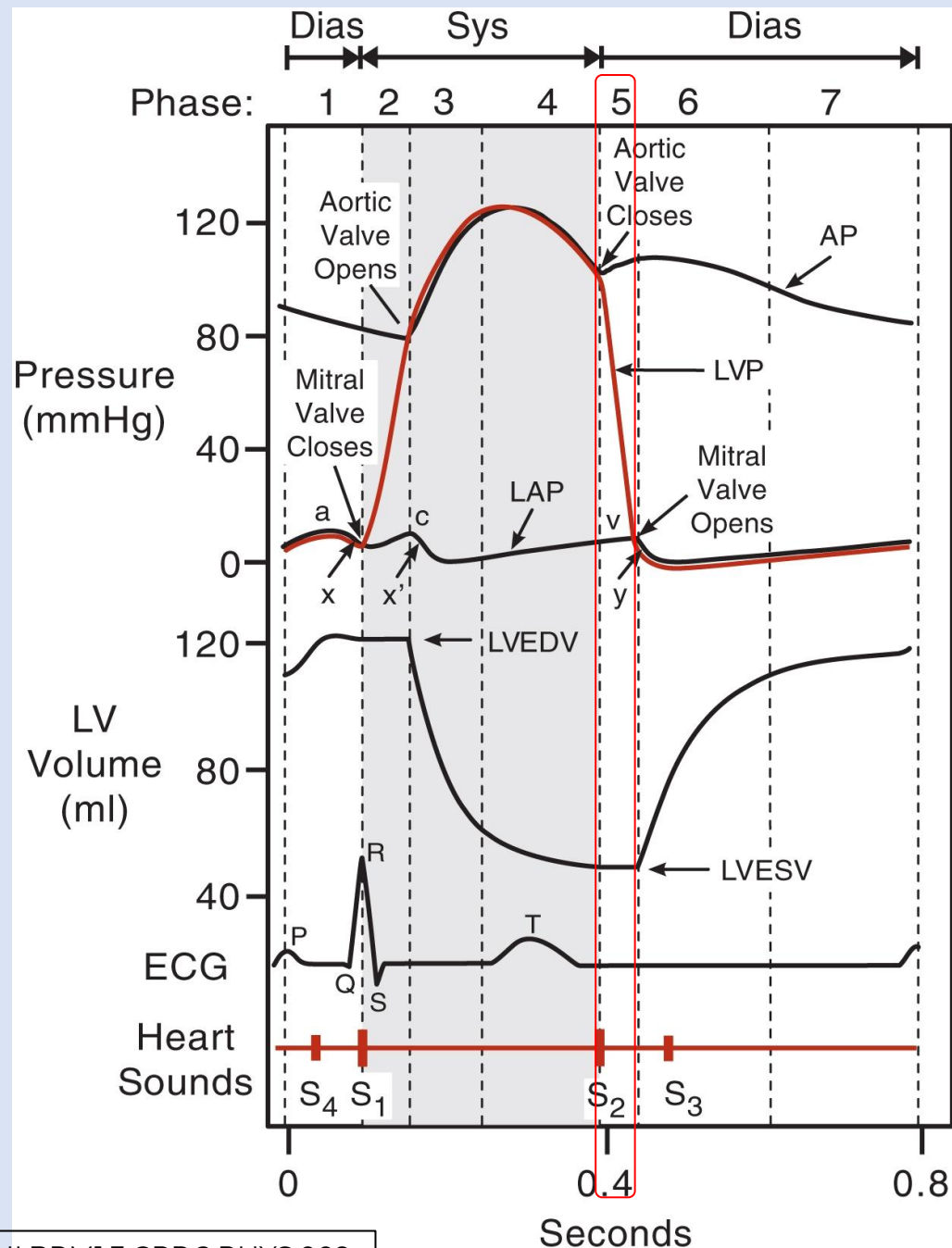
3. Rapid ventricular ejection

4. Reduced ventricular ejection

70% of stroke volume is pumped during the first third of the ventricular ejection and therefore is called **PERIOD OF RAPID VENTRICULAR EJECTION**.

Remaining 30% of blood gets pumped during the next two thirds of ejection and is called the **PERIOD OF REDUCED VENTRICULAR EJECTION**.

Emptying of ventricle reduces ventricular volume to **ESV**. Blood getting into aorta rises aortic pressure to the peak and then slowly decreases as the flow becomes weak. At the end of this phase, aortic valve closes producing S2 sound.



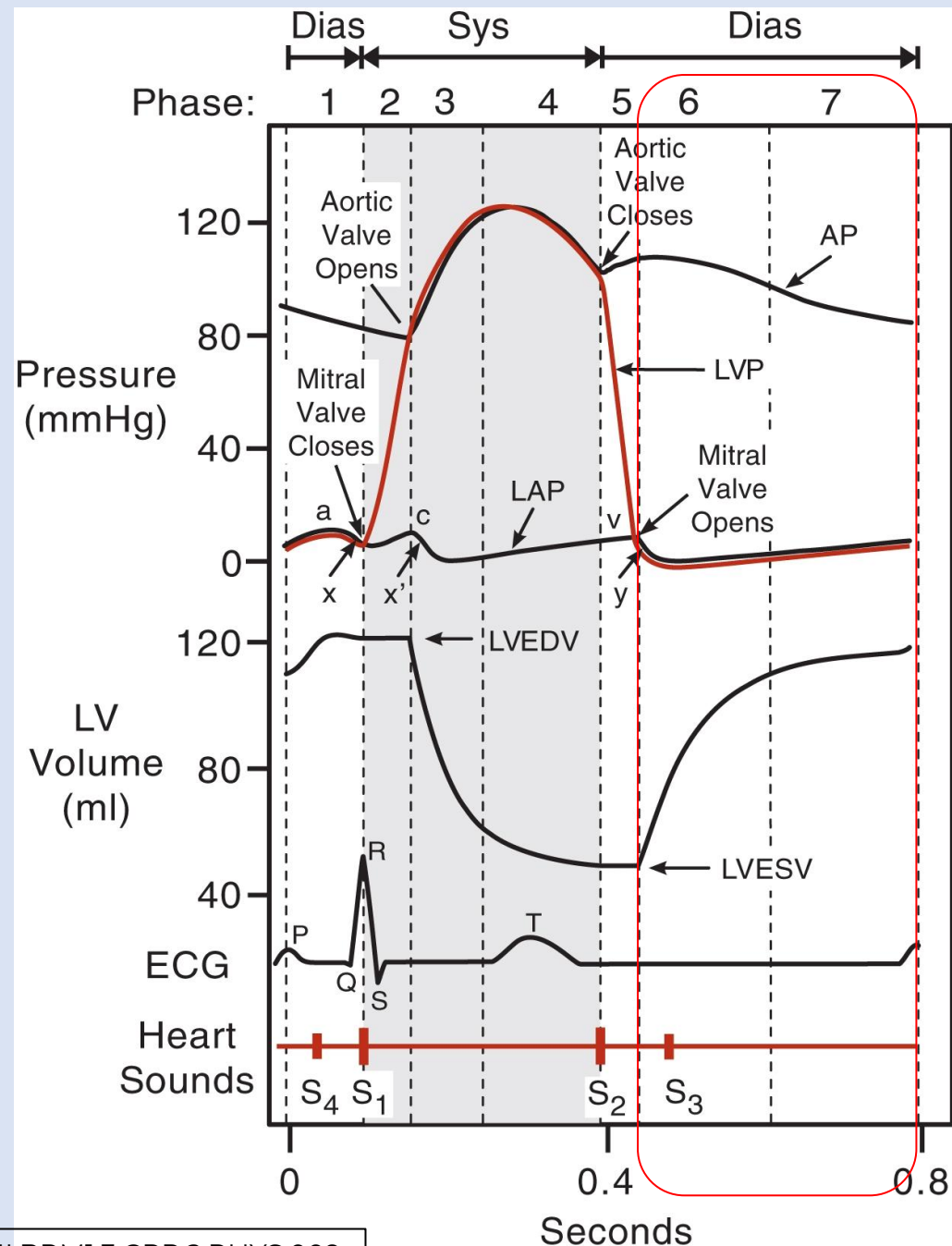
5. ISOVOLUMETRIC VENTRICULAR RELAXATION:

This is the period between aortic valve closing and mitral valve opening. ALL VALVES CLOSED at this time.

The ventricle begins to relax as a closed chamber and therefore, the **intraventricular pressure drops** rapidly as shown in LVP tracing.

Since both valves are closed, ventricular volume remains same as **ESV** (hence the name isovolumetric relaxation).

Continued filling of the left atrium makes the left atrial pressure to reach maximum and is represented by **"v" wave** in LAP tracing.



6 & 7. VENTRICULAR FILLING PHASES:

It occurs at 2 phases:

6. Rapid ventricular filling

7. Reduced ventricular filling.

As the left ventricular pressure is lower than that in left atrium, mitral valve opens and the ventricular filling starts.

6. Rapid ventricular filling:

Begins as soon as the mitral valve opens. During this phase, left ventricular pressure continues to decline because ventricular relaxation occurs more rapidly than does the filling of the ventricle. The first part of the filling is rapid with large volume of blood getting into the ventricle because of the pressure gradient. Ventricular volume increases.

Rapid filling produces **S₃**. Atrial blood emptying into ventricle reduces atrial pressure and is indicated by a **"y" wave** in LAP.

7. Reduced ventricular filling (diastasis): The period during which ventricle continues to relax but fills a little. Intraventricular pressure remains fairly low and constant during this

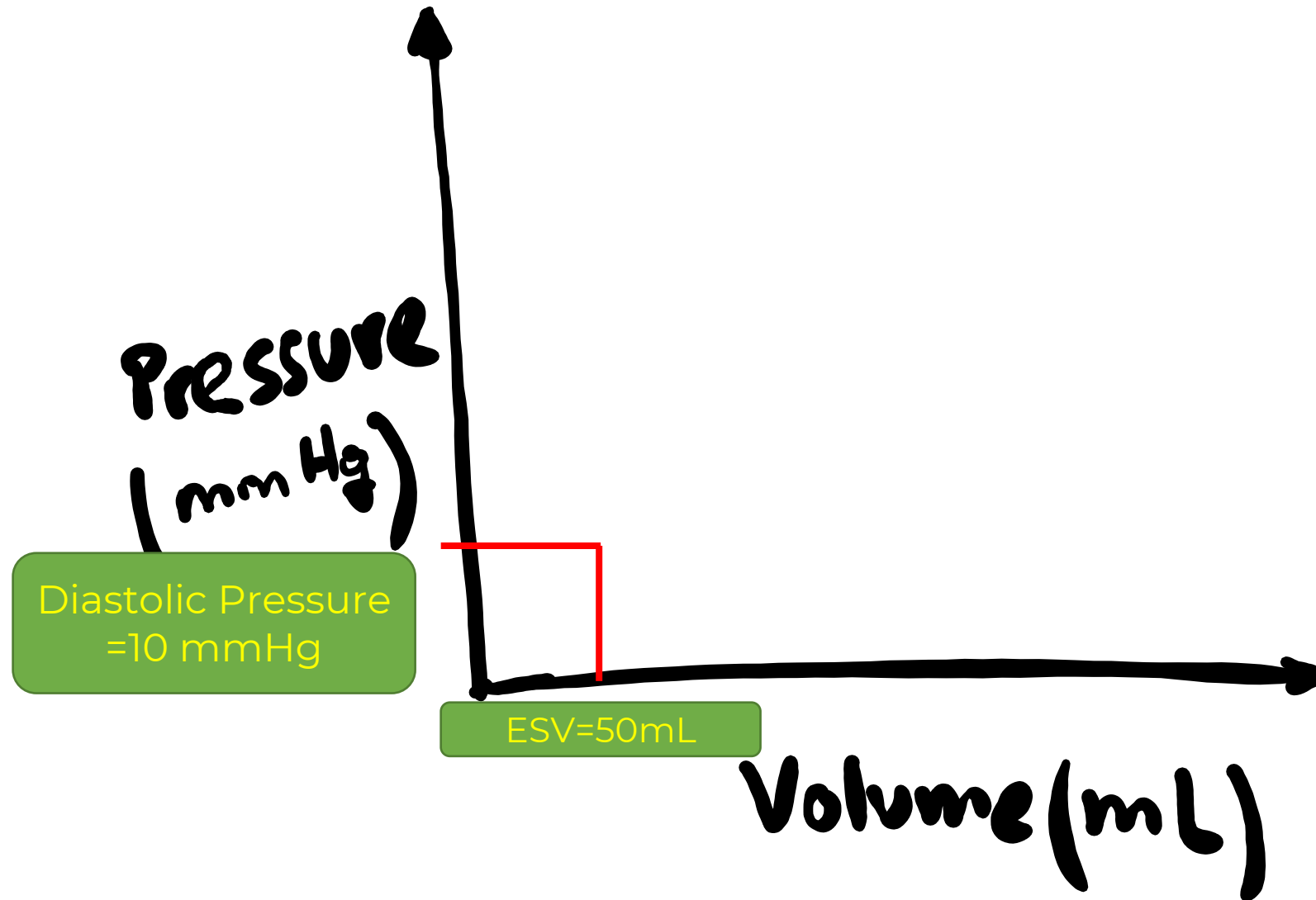
Lecture Outline

- ❖ Cardiac Cycle Phases
- ❖ Graphic representation of changes in pressure, volume & sounds
- ❖ **Left Ventricular Pressure-Volume Loop**
- ❖ Left side versus Right side of the heart
- ❖ Turbulent Flow
- ❖ Heart Sounds
- ❖ Heart Murmurs

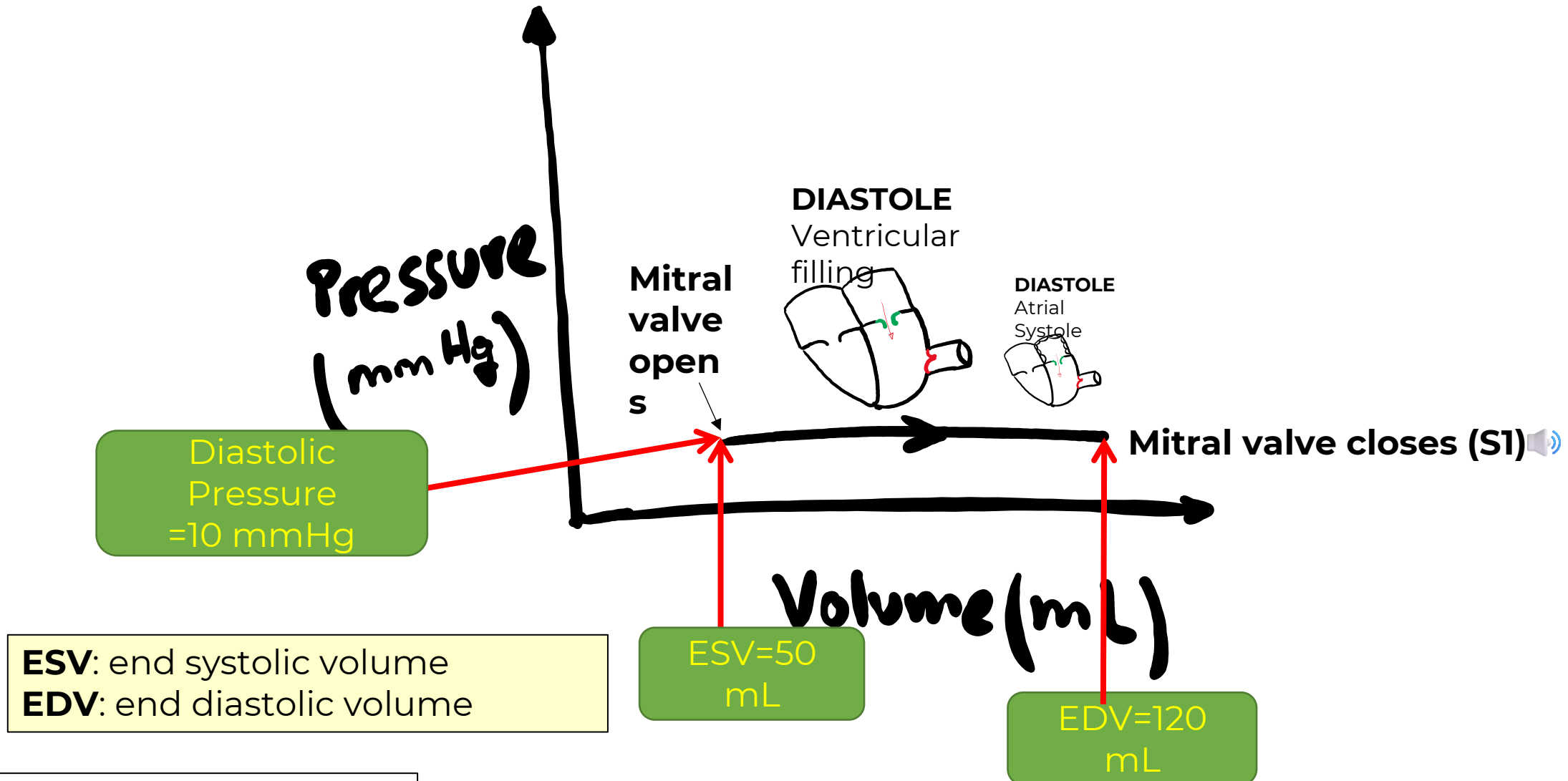
Ventricular Pressure-Volume Loop

- Representation of changes in **ventricular pressure** in relation to its **volume** during one cardiac cycle.
- Pressure against volume plot (does NOT show time!)
- Reflects ventricular function during ejection and filling (cardiac work)
- Useful to interpret ventricular functioning under physiological and diseased conditions

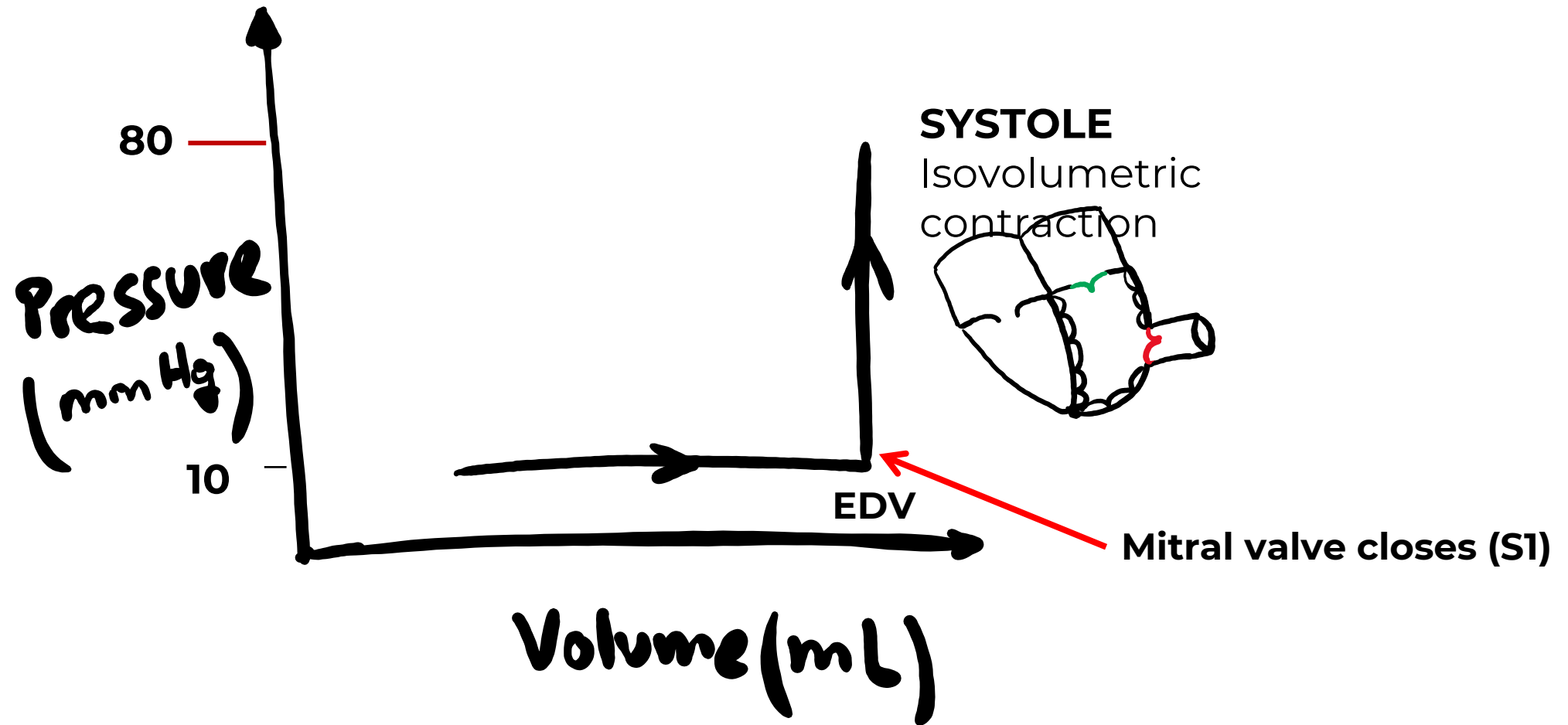
Ventricular Pressure-Volume Loop



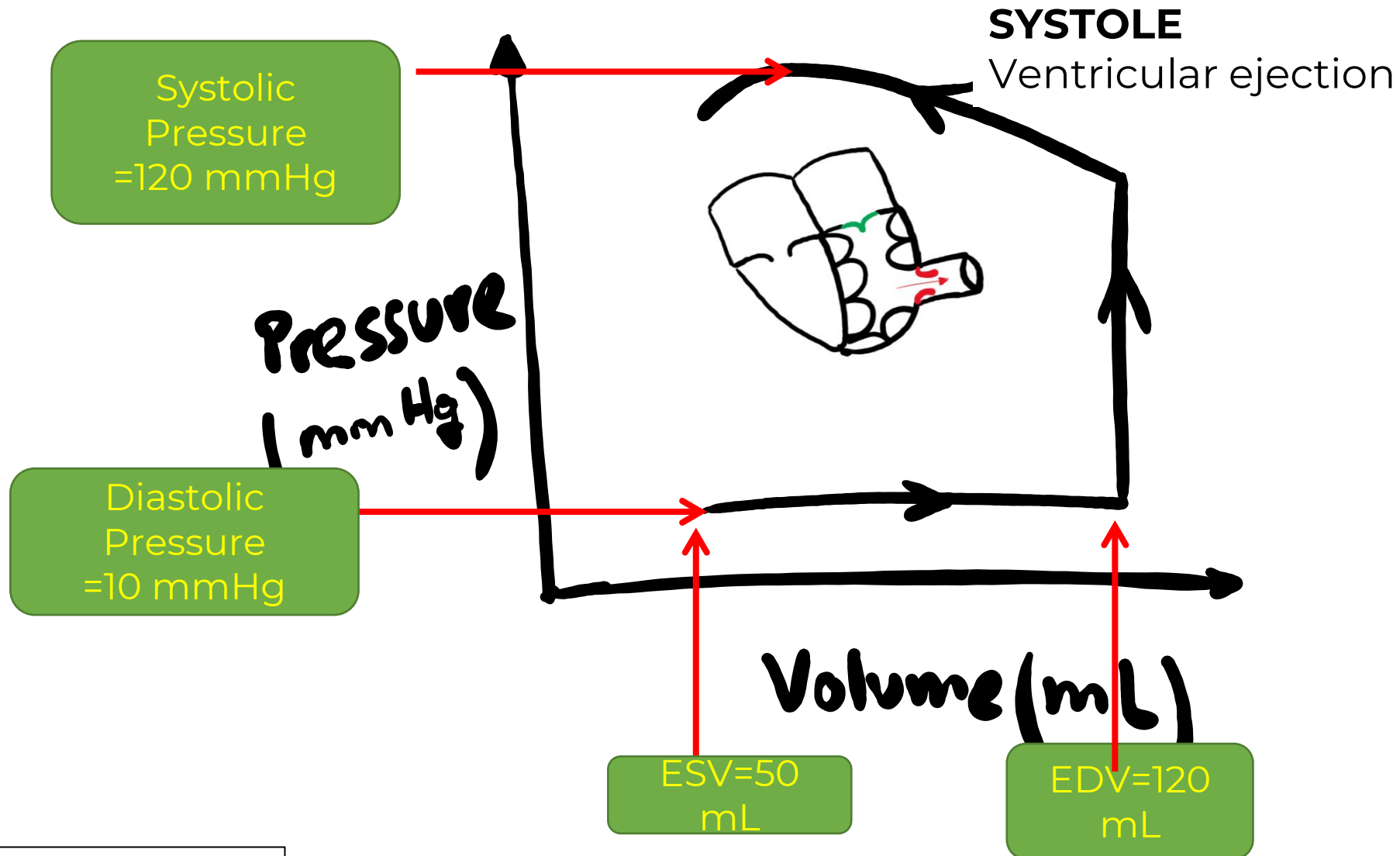
Ventricular Pressure-Volume Loop



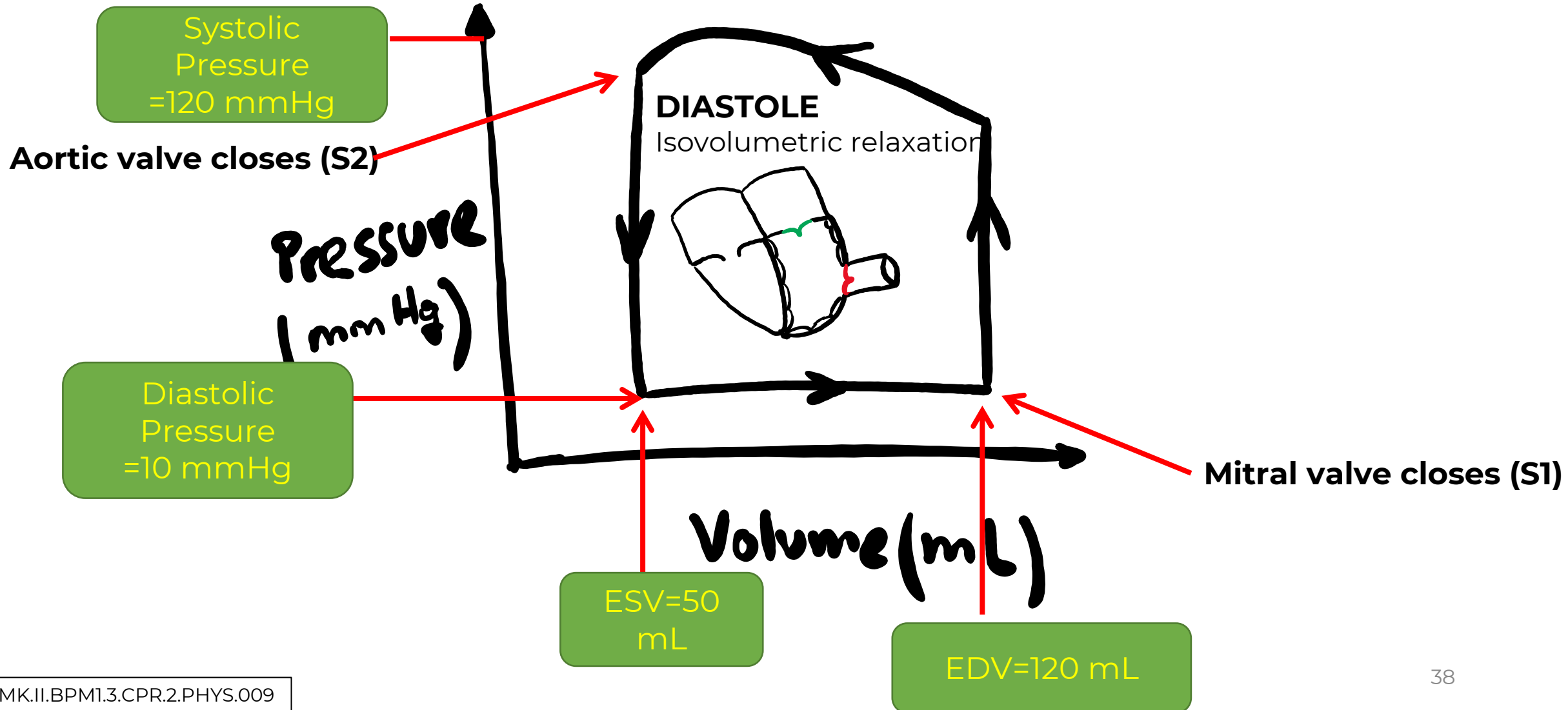
Ventricular Pressure-Volume Loop



Ventricular Pressure-Volume Loop



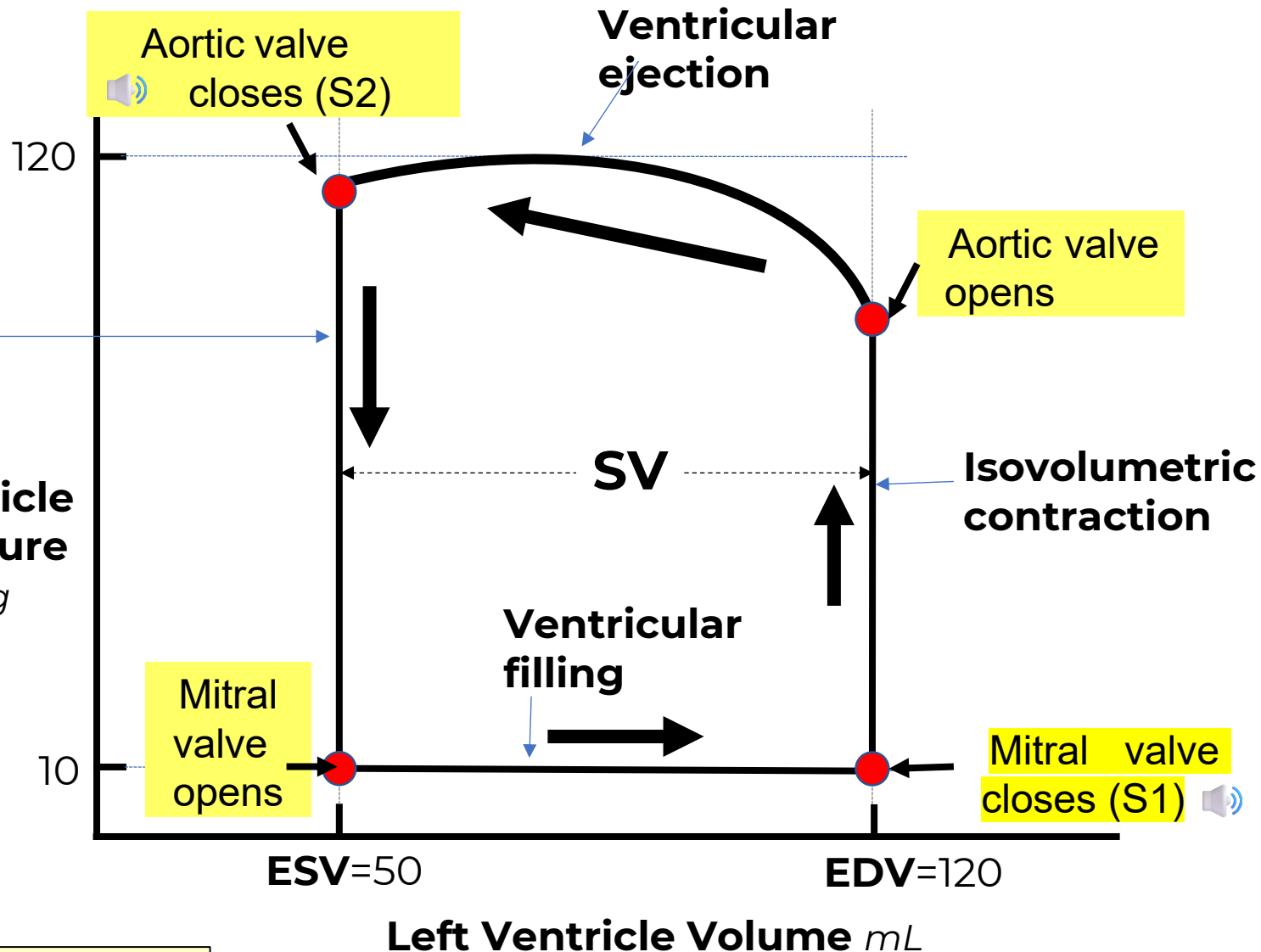
Ventricular Pressure-Volume Loop



Isovolumetric relaxation

The area enclosed by the PV loop represents the left ventricular stroke work

Left Ventricle Pressure
mmHg



ESV: end systolic volume
EDV: end diastolic volume
SV: stroke volume; **$SV = EDV - ESV$**

Ejection fraction of the left ventricle
 SV/EDV , normal value > 0.55 (55%)

Lecture Outline

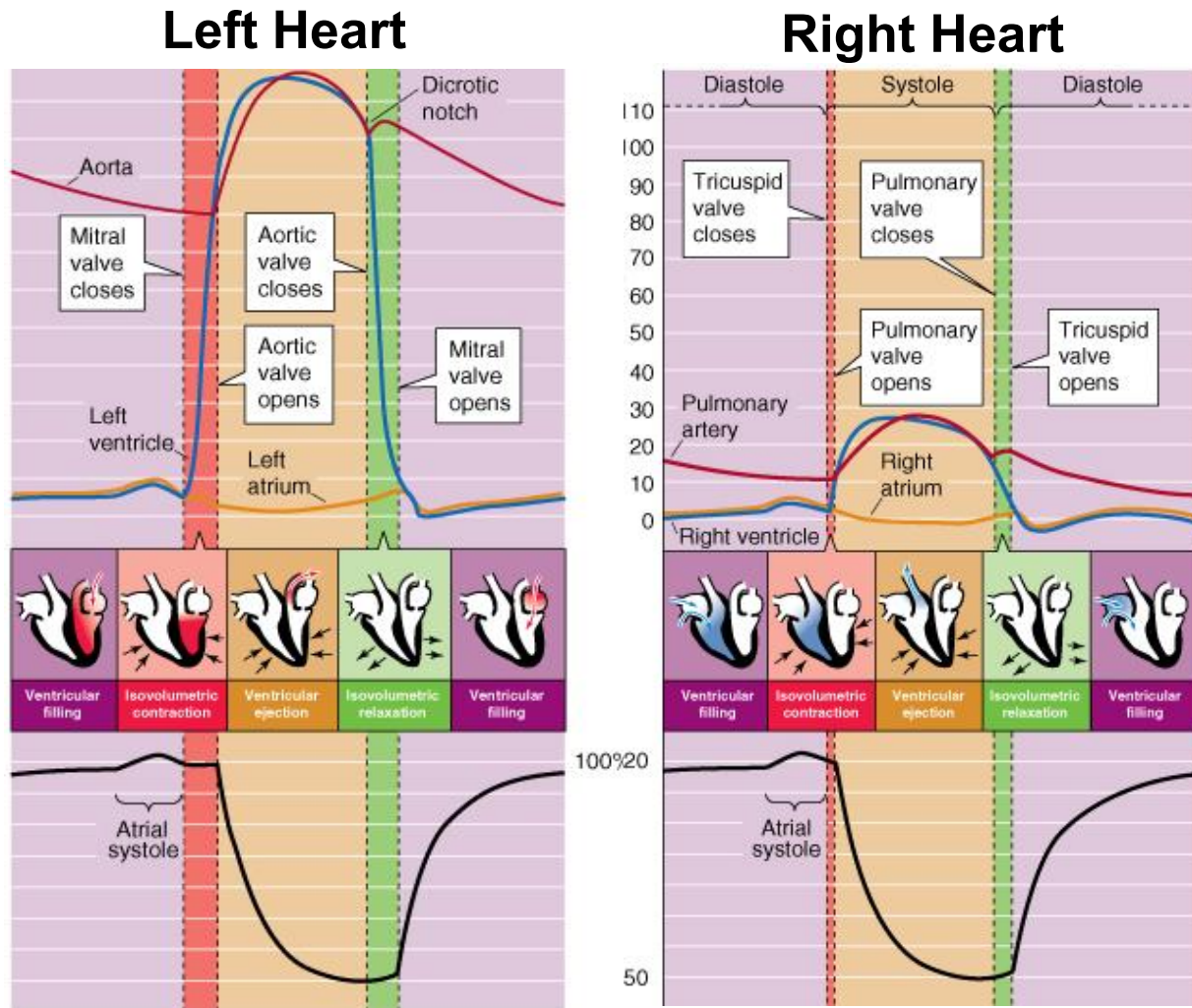
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CARDIAC CYCLE: Left side vs Right side

Similarities:

- Because the functional syncytium, similar mechanical events occur **simultaneously** in both the left and right ventricles/atria.
- The valves of the right and left sides of the heart open and close simultaneously with a *slight delay on the **right** side*.
- Both ventricles pump the same amount of blood and therefore must have identical stroke volumes and cardiac output.

CARDIAC CYCLE: Left side vs Right side



The **major difference** between the right and left side is the **pressure values**.

The right side operates at much lower pressures because the **pulmonary system is a low resistance circulation**.

Pressures generated in the RV are much less than in the LV

Normal Cardiac Pressures: Right vs Left side of the heart

STRUCTURE	PRESSURES (mm Hg)
Left atrium	< 12
Left ventricle	120/10
Aorta	120/80
Right atrium	< 5
Right ventricle	25/5
Pulmonary artery	25/10

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Heart Sounds: Genesis, Timing & Audibility

Heart sounds	Genesis & Timing	Audibility
S1 (M1T1)	Occurs during mitral and tricuspid valve closure. Sound results from the vibrations of blood and the chordae tendineae in the ventricles.	Normally audible
S2 (A2P2)	Occurs during aortic and pulmonary valve closure. Sound results from the vibration of the valves and aortic wall.	Normally audible
S3	Occurs in early diastole during rapid ventricular filling phase. Sound is originated by ventricular filling.	not normally audible; may be audible in children
S4	Occurs in late diastole when atrial systole fills ventricle ("atrial kick"). Sound is originated by ventricular filling.	Not normally audible; if audible, indicates disease state

Gallop rhythms

S3 and S4 sounds are not normally audible. When they are loud and audible, they occur along with S₁ and S₂. The resulting combination of heart sounds is called **gallop rhythm** (resembling the sound of a galloping horse). If S3 becomes loud, it occurs shortly after S₂ during the period of rapid passive ventricular filling. This combination of three heart sounds is called **ventricular gallop rhythm**. Loud S3 is an important physical examination finding in patients with systolic left ventricular failure. If S4 becomes loud, it occurs shortly before S₁ during atrial systole (last part of ventricular diastole). Thus, the combination of S₁, S₂, and S₄ sounds is called an **atrial gallop rhythm**. An audible S₄ indicates that the ventricle is stiff, a physical finding in patients with a stiff ventricle.

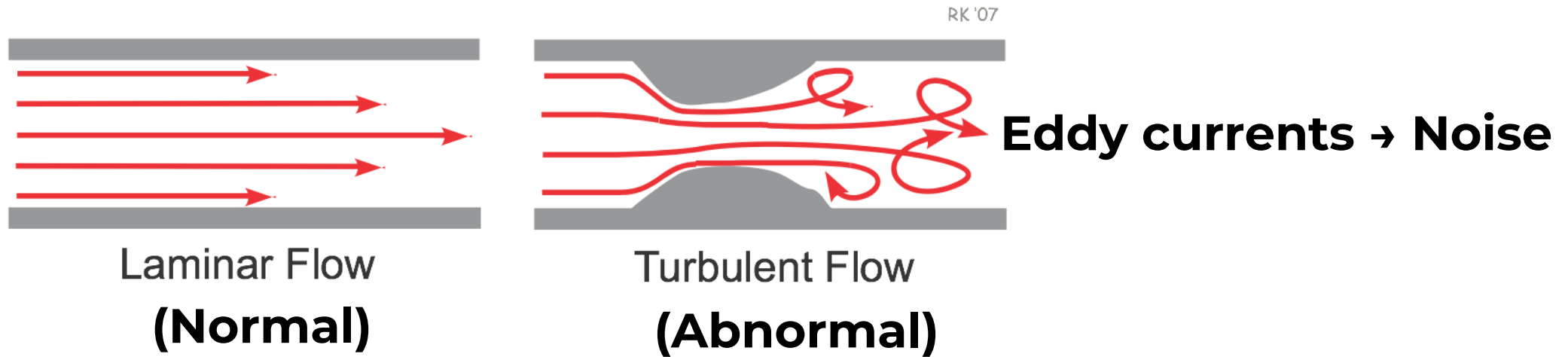
Splitting patterns of the second heart sound.

Normal splitting PHYSIOLOGIC	Inspiration produce a drop in intrathoracic pressure , increasing venous return, increasing right ventricular filling, increasing right ventricular stroke volume, which produces a delayed closure of the pulmonic valve.
Wide splitting	Conditions that delay RV emptying: • Pulmonic stenosis • Right bundle branch block
Fixed splitting	Condition: • Atrial septal defect: left-to-right shunt increases right atrium and right ventricle volumes, increasing flow through pulmonic valve which produce a delay in the pulmonic valve closure during both inspiration and expiration. Thus, the split is fixed (does not change with phases of breathing).
Paradoxical splitting	Normal order of valve closure is reversed (P2 occurs before A2). This is because aortic valve closure is delayed. Conditions: • Severe Aortic stenosis • Left bundle branch block (In the above patients, during inspiration, P2 also closes later and moves closer to A2, paradoxically eliminating the split usually heard in expiration).

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Turbulent Flow and Factors contributing



- Factors contributing: Velocity of flow, Diameter, Blood viscosity and density.
- Turbulence is measured by Reynolds number. A flow is turbulent if the number is more than 2000.
- Turbulence generates audible vibrations in heart and blood vessels that can be heard with a stethoscope (e.g., [cardiac murmurs](#), [carotid bruits](#)).

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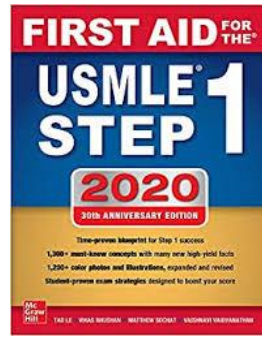
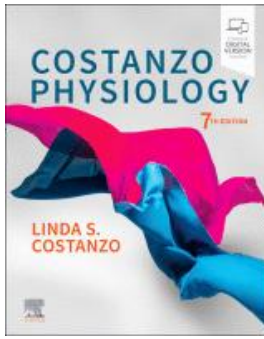
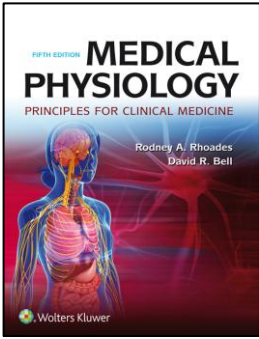
Heart Murmurs: Genesis and Types

- Heart murmurs are abnormal sounds that occur during systole or diastole.
- They appear when blood flow through valves becomes abnormal because of **stenosis or leakage of the valves.**
- They also appear when abnormal flow occurs because of anatomical defects
- Most common physical findings of **valvular heart diseases**
- Three types: **Systolic**, **Diastolic** and **Continuous murmurs.**

MURMURS AND HEART DISEASES

SYSTOLIC MURMURS	DIASTOLIC MURMURS	CONTINUOUS MURMUR
Aortic & pulmonary stenosis	Aortic regurgitation	Patent ductus arteriosus
Mitral & tricuspid regurgitation	Mitral stenosis	
Mitral valve prolapse <i>MOST FREQUENT VALVULAR LESION</i> <i>A late-systolic murmur (murmur occurs in the later part of systole)</i>	Tricuspid stenosis	
	Pulmonary regurgitation	
Ventricular septal defect		

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Do you have questions?

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